



MASTER'S THESIS: PUBLIC POLICY AND
DEVELOPMENT

**A Giant with Feet of Clay: Public
Reinsurance and Climate Change**

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August 2024

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Abstract

As climate risks intensify, understanding how insurance frameworks cope with uninsurable threats becomes increasingly critical. This study investigates the interplay between insurance and climate change by examining France's natural disaster compensation system (Cat-Nat), a prime example of how government-backed disaster insurance influences risk perception and property pricing. Focusing on the systemic risks posed by the shrinkage and swelling of clay soils, we use a quasi-experimental design to reveal the potential market consequences of excluding these risks from the insurance framework. Our findings not only underscore the Cat-Nat system's effectiveness in mitigating financial risks for vulnerable homeowners but also raises significant concerns about its sustainability in the face of escalating climate-related challenges. This study offers valuable insights into broader discussions on insurance and climate change, illustrating how market mechanisms respond to emerging, uninsurable risks.

Keywords: climate change, insurance, uninsurable risks, property markets, Cat-Nat system

JEL codes: Q54, G22, R31

Code is provided in the GitHub repository (<https://github.com/Sacha-N/Shrink-swell-public-repo.git>). I would like to thank Mathias André, Inès Moutachaker, and Julie Sixou for their warm reception at Insee and for their guidance.

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1 Introduction

In-pricing of climate change in insurance policies is seldom explicit, yet a worldwide trend of rising premiums and reduced insurance coverage has been identified (Khoo and Yong 2023). Very exposed territories may become "too hot to insure", as it has been observed in the wake of the 2018 Californian megafires (Walsh 2018). The economic literature has promptly highlighted that rising insurance premiums or the inability to secure coverage might incentivize risk-averse individuals to reduce their exposure to climate risks by relocating to less vulnerable areas (Kousky 2019).

However, the efficacy of such market mechanisms hinges on the clear perception of risk. It is broadly acknowledged that agents often underestimate the risks associated with natural disasters (McCoy, McDonough, and Tra 2024, Pautard 2023). The crossing of private insurance tipping points, which may trigger a sharp rise in salience, occurs when the essential conditions for sustainable insurance are not satisfied: namely, the existence of insurable risks, an insurable population, and solvent insurers (Stephen J. Collier and Lehtonen 2021).

In situations where a risk is deemed uninsurable, adverse selection may indeed occur, leading to the potential collapse of the insurance market (Stephen J. Collier and Lehtonen 2021). Consequently, the increasing frequency and geographic dispersion of extreme climatic events often comes with greater public intervention, such as disaster funds, climate reinsurance programs, or mandatory insurance schemes. In the case of floods, the most common and destructive type of natural disaster, national configurations vary greatly (Lamond and Penning-Rowsell 2014): most developing countries have no public or mandatory private scheme, leading to very low coverage, while few developed countries have close to complete coverage, thanks to mandatory private insurance (Iceland, Norway), public-private insurance linked to home insurance (France) or general public schemes (Spain). These different national configurations are the result of a trade-off: How can we reconcile the population's need for protec-

tion with the financial sustainability of an increasingly expensive system? (Lucas and Booth 2020)

In the literature, the significance of natural risks is commonly assessed by examining their impact on property prices and interpreting the observed premiums in light of insurance practices (Zhang and Leonard 2018). This study, however, takes a different approach by investigating the extent to which major risks can be overlooked by economic agents, particularly as a consequence of the insurance mechanism. Moreover, at a time when climate insurance is under threat, this study explores the potential consequences of insurance system failures.

France’s natural disaster insurance scheme offers a distinctive case study for evaluating the pricing of uninsurable risks, which are indirectly covered by government-backed insurance. The French model operates through the Caisse Centrale de Réassurance (CCR), a public firm. Disaster coverage is mandatory and included in property insurance, with a flat-rate premium surcharge collected by direct insurers. This approach ensures a broad insurable population, while the flat-rate cross-subsidization enhances affordability. However, the system produces a muted price signal for risk, which may obscure systemic threats and complicate risk assessment and management. This contrasts with models in other countries that have different public-private configurations or non-mandatory insurance systems.

A critical aspect of the systemic threats challenging France’s climate risk insurance system is the risk associated with differential soil movement caused by the shrinkage and swelling of clays (Cour des comptes 2022). Clay soils undergo volumetric changes depending on their water content (Barthelemy et al. 2024). Droughts induce shrinkage, while water infiltration causes swelling, compromising the structural integrity of buildings, especially those with shallow foundations. Drought episodes, all the more when they recur, can then cause the appearance of cracks in buildings, predominantly individual houses which are more vulnerable than collective buildings (Ledoux 2023).

Over the past 35 years, the shrink-swell behavior of clay has become the second-largest compensation item under the Cat-Nat regime, rising to the top position in 2022. The increasing frequency of severe droughts since 2017 has drawn attention to the severe threat that clay risk poses to the natural disaster compensation system, prompting the release of several institutional reports¹. As discussions on improving the system continue, evaluating its current performance and impact is crucial.

This study makes three key contributions. First, it demonstrates that the risks associated with clay shrinkage and swelling are only beginning to be reflected in property prices, despite the significant and exponentially growing risk they pose to buildings. The historically low visibility of this risk, which is expected to increase with climate change, may partly be explained by the Cat-Nat system. Currently, natural disasters do not appear to serve as a strong risk signal in France. However, using a quasi-experimental design, this study shows that the refusal of insurance coverage in a municipality affected by clay shrinkage and swelling has a substantial impact on the property market. This suggests that removing clay risk from the Cat-Nat system, as sometimes discussed (Cour des comptes 2022), would have profound consequences. Ultimately, the findings indicate that the Cat-Nat system is particularly effective, and perhaps too big to fail, raising critical questions about the financial burden borne by the nation to compensate for vulnerable constructions located in risk-prone areas, especially when alternative measures are available.

The first section of this paper outlines the institutional framework, tracing the history and operation of the public-private reinsurance system for natural disasters. Following this, the data sources are presented: these include data on soil composition and on droughts to document current and future risk, administrative recognition data to provide insights into the insurance mechanism, and outcomes related to housing,

¹These include a report by the Court of Auditors (Cour des comptes 2022), a synthesis by France Assureurs (France Assureurs 2022), an information report by the National Assembly (Marsaud and Rousseau 2023), and a parliamentary mission report (Ledoux 2023).

its inhabitants, and property prices. The third section details the empirical setup: initially, the pricing of clay shrinkage and swelling risk is evaluated using a difference-in-difference approach, followed by an assessment of the significance of insurance using a regression discontinuity design (RDD). The fourth section presents the results, and the fifth discusses the underlying mechanisms and limitations.

2 Institutional Background

The French Cat-Nat system, established in 1982, is a public-private insurance mechanism designed to mitigate the financial impact of natural disasters on property owners. This system mandates that all property insurance policies include coverage for natural disasters, ensuring broad protection². The process begins with the government officially recognizing a natural disaster in selected municipalities through a decree, which triggers the Cat-Nat coverage. Policyholders then have a specified period to report their claims to their insurers, who assess the damages and provide compensation accordingly. This system is financed through a surcharge on home and car insurance premiums, thereby spreading the risk and cost across a large pool of policyholders, enhancing financial resilience against natural disasters.

In response to numerous droughts in the latter half of the 1980s, damages on houses caused by shrink-swell were incorporated into the Cat-Nat regime in 1989, making France the only European country to cover this risk within the framework of a public insurance system (Cour des comptes 2022). Since then, drought risk has accounted for 36% of Cat-Nat claims (1989-2019, CCR), and has risen to become the largest burden on the public-private framework in the past ten years (CCR 2024).

Households' residences are covered by the Cat-Nat guarantee; however, not all assets of local communities are covered. For instance, roadways can be affected by shrink-swell but are not subject to compensation. Given that home repairs are often

²99% of households in metropolitan France have their homes insured, whereas only 52% are insured in overseas departments (Gislain-Letrémy and Calvet 2011).

substantial, the average cost of a shrink-swell claim amounts to €16,300 (Cour des comptes 2022), making it the highest individual Cat-Nat damage cost. At the aggregate level, the cost of claims for the 2022 drought ranges between €3 billion and €3.5 billion (CCR 2024), a record since the establishment of the compensation regime. This sharp increase in compensated claims explains the rise in the Cat-Nat surcharge from 12% to 20% on multi-risk home insurance (MRH) policies as of January 1, 2025 (Sénat 2024).

The Cat-Nat system is designed to cover extreme, uninsurable risks where a causal link between a climatic event and damages can be demonstrated. However, in the case of shrink-swell clay phenomena, establishing this causal relationship is notably challenging (Barthelemy et al. 2024). Shrink-swell is a multifactorial, gradual phenomenon with a slow progression (for an in-depth explanation, see Appendix 9.1). For instance, the appearance of cracks in a house in June could result from an abnormal spring drought recognised by decree, or from a summer drought that was not officially recognised. This ambiguity in pinpointing the exact cause complicates the process of assigning and validating claims under the Cat-Nat system (Cour des comptes 2022). In practice, two criteria are considered for coverage under the Cat-Nat guarantee:

- A geotechnical criterion: The Bureau of Geological and Mining Research (BRGM) verifies the presence of clay layers in the municipal soil. At least 3% of the municipality's surface area must consist of clay patches.
- A meteorological criterion: To assess soil dryness, Météo France calculates the variation in the Soil Wetness Index (SWI) of superficial soils using an 8 km by 8 km grid. When this index surpasses a specific threshold, Météo France classifies the drought as abnormal for all municipalities within the grid. For any event occurring between January 1, 2018, and December 31, 2023, a drought is considered exceptional when the Soil Wetness Index has a return period of at least 25 years. Practically, this means that the SWI for a given month must be

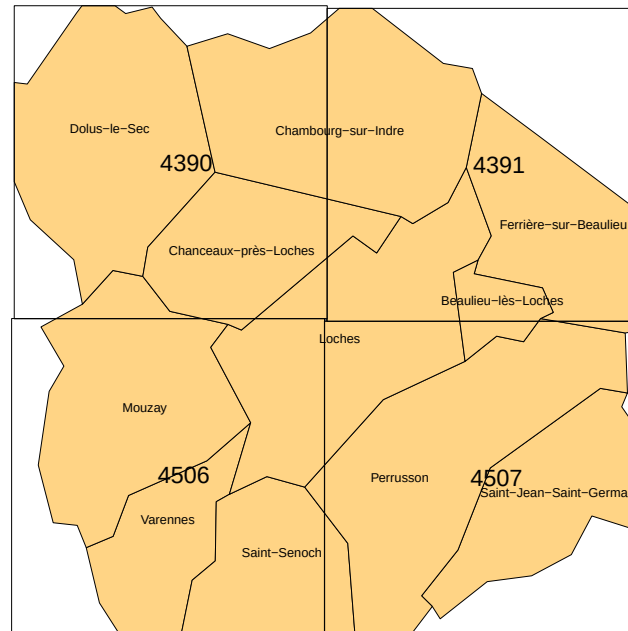
among the two lowest recorded in the past 50 years for the same month.

While the geotechnical criterion is unambiguous, the meteorological criterion has led to dissatisfaction and misunderstandings (Sénat 2023). Half of the municipalities that requested recognition of the state of natural disaster during the 2019-2020 droughts did not obtain it (Cour des comptes 2022). It is common for two neighbouring municipalities to experience similar shrink-swell damage, yet one is recognised as a natural disaster while the other is not. It is sufficient for a part of the municipal territory to be located within a grid cell meeting the meteorological criterion for the entire municipality to be eligible. Figure 1, replicated from Circular n°INTE1911312C 2019, which defines the meteorological criterion, illustrates border effects: the criterion for the municipality of Mouzay can be valid thanks to grid 4390, and invalid for Varennes, even though both municipalities have the larger part of their surface in grid 4506. This imprecision is a source of litigation: according to the Cour des comptes 2022, 93% of litigation concerning recognition orders of municipalities in natural disaster states relate to shrink-swell.³

Lastly, even within municipalities recognised as natural disaster zones, a significant portion of compensation claims are denied by insurance company-appointed experts.

Reports from the Court of Auditors, the Senate, and the parliamentary mission call for a rapid improvement of the meteorological criterion, which has been addressed with the Circular n°IOME2322937C 2024 issued on April 29, 2024. The threshold for qualifying an exceptional drought based on the Soil Wetness Index (SWI) has been lowered from a return period of 25 years to 10 years for all events occurring from January 1, 2024, onwards. Furthermore, the repetition of droughts with a return period of at least 5 years will now be sufficient to meet the meteorological criterion. The underlying rationale is that the cumulative impact of repeated lower-intensity events can be more damaging than a single, isolated exceptional drought (Circular

³As of December 31, 2020, this involved 924 disputes, which is few considering the number of municipalities whose request for natural disaster recognition was rejected.



Source: Safran grids (Météo France, 2021), replication of the figure from the circular n°INTE1911312C (May 2019)

Figure 1: Border effects in the Cat-Nat meteorological criterion

The example of Loches (Indre-et-Loire)

n°IOME2322937C 2024). Additionally, to mitigate the border effect, a municipality with a return period of 5 years adjacent to a recognised municipality will now be granted recognition. Overall, this new circular relaxes the criteria used to assess clay-related natural disasters, aiming to reduce the number of false negatives, i.e., instances where municipalities are not recognised as being in a state of natural disaster despite drought-induced shrink-swell effects.

The relaxation of the meteorological criterion is expected to reduce the proportion of homeowners experiencing damage without compensation, although it may also increase the financial burden on the public reinsurance scheme. Since 2017, the Cat-Nat

regime has been operating at a deficit⁴. Consequently, reforms aimed at removing shrink-swell risk from the Cat-Nat guarantee are under discussion, although no significant changes are anticipated in the medium term. According to the Sénat 2023, a private insurance framework would lead to prohibitive premiums and significantly lower coverage rates, while the establishment of a specific public regime would be highly costly. However, maintaining the current regime is unsustainable in light of the projected increase in drought episodes. During a hearing for Sénat 2024, CCR indicated that recent droughts have severely depleted its reserves, which are expected to approach zero by 2024; the planned increase in the Cat-Nat surcharge on January 1, 2025, will be insufficient to address this issue.

3 Data

3.1 Climate data

Claims data collected by insurers is seldom accessible to researchers, and to date, no geolocated projections of clay-related subsidence damage related to climate change have been produced. As part of our work at INSEE, we have developed an innovative strategy to create a geolocated proxy for shrink-swell damage, applicable to both historical data and future projections. Drawing inspiration from the Cat-Nat criterion, which requires the concurrent presence of clay soils and abnormal droughts to establish a causal link between climate data and observed damages, we established a climatic database that identifies dwellings at risk over time. To achieve this, we cross-referenced two available datasets: the shrink-swell exposure map created by the Bureau of Geological and Mining Research (BRGM), and the projections of surface soil drought indicators (SWI) produced by Météo France as part of the Drias project.

3.1.1 Map of clay shrinkage and swelling exposure

National studies of soil composition were conducted by the BRGM between 1997 and 2010. Based on this initial zoning, referred to as susceptibility zoning, 24% of the

⁴However, CCR has only needed to invoke state guarantees once, during the 1999 storm. These recent deficits have therefore resulted in a decrease in CCR's reserves.

metropolitan territory was classified as having a medium or high level of susceptibility to shrink-swell in 2010. Susceptibility is determined using three criteria: lithological characteristics (proportion of clay minerals, i.e., particles smaller than 2 micrometers), mineralogical composition (proportion of swelling minerals), and geotechnical properties (plasticity).

In subsequent years, it became evident that numerous municipalities located in zones classified as having low susceptibility experienced significant levels of damage. As a result, the BRGM maps were updated in 2019-2020 under the framework of the ELAN law by incorporating loss data collected by the insurance sector in the Sinistres Indemnisés Liés aux Évènements Climatiques (SILECC) database, which compiles historical loss data from the Cat-Nat regime. This cross-referencing enabled the BRGM to assign local hazard exposure levels: non-clay zones, and zones of low, medium, and high exposure (map in Appendix 9.2). The exposure level is determined using a cross-referencing table: medium susceptibility combined with low or medium loss corresponds to medium exposure, while high susceptibility corresponds to high exposure, regardless of past losses.

As a result, the proportion of metropolitan territory with medium or high exposure has increased to 48%, now accounting for 93% of observed losses (BRGM, 2020). The evolution of loss data thus explains the differences in the mapped hazard, as the foundational geological maps remain the susceptibility maps developed by the BRGM between 1997 and 2010.

In 2021, the Statistical Data and Studies Department (SDES) of the Ministry of Ecological Transition conducted a matching of Fidéli, an exhaustive dwelling database, with the resulting exposure map (Antoni 2021). This matching identified 10.4 million individual houses with medium or high exposure, representing 54.2% of the metropolitan total. This figure shows that shrink-swell threatens the majority of homeowners, but does not provide information on the evolution over time and the intensity of

possible damage.

The geological maps created by the BRGM assess the presence of clay formations at a 1/50,000 scale. The granularity of the susceptibility map implies that some small clay areas may not be included in the ELAN zoning, and conversely, certain parcels within zones marked as susceptible to shrink-swell may actually consist of non-clay or non-sensitive terrain. This underscores the need for thorough soil studies before construction. Therefore, the exposure maps may not always be accurate at the parcel level.

3.1.2 Soil Wetness Index (SWI)

Clay soils are a predisposing factor for shrink-swell, but in most cases it is droughts that trigger the phenomenon. To quantify this, we utilize the Soil Wetness Index (SWI), which characterizes the state of water reserves relative to the available water capacity at a depth of approximately 2 meters, essentially the water available for plant uptake. Since there are no superficial soil wetness sensors in each municipality, the index values are derived from the SIM model, which is part of a family of hydro-meteorological models from Météo-France designed to model the water cycle. These models incorporate local topography and vegetation, based on a set of hypotheses tailored to each setting. The SIM model divides the territory of metropolitan France into geographical grids of 8 kilometers per side, covering a total of 8,981 grids. Each grid, hereafter Safran grid, is numbered and encompasses all or part of a municipality. An SWI value close to or above 1 indicates very wet soil, while an SWI close to or below 0 indicates very dry soil. The Safran grid system is fixed and does not change from year to year.

When a municipality requests recognition of a natural disaster, Météo-France produces a report assessing the meteorological criterion based on the uniform SWI values from the Safran grids intersecting the municipality's surface. Uniform SWIs are calculated using a specific method⁵ and a set of assumptions that ensure equal treat-

⁵The uniform SWI is defined as the average of the daily SWIs for the current month and the

ment of natural disaster recognition requests across the nation⁶. While this report is not publicly available, historical uniform Soil Wetness Index values covering every month since January 1969 are accessible as open data, allowing for the replication of the assessment.

In addition to the output from Météo-France models using observed data, the ClimSec project by Météo-France, developed between 2008 and 2011, aimed to model the evolution of water reserves in the superficial layers of soils over a reference period (1958-2008) and project up to 2100 (Météo France 2011). The prospective component is based on regionalized models derived from IPCC scenarios⁷. The increase in drought episodes affects the entire territory, particularly the northern and northeastern regions, which were previously less impacted. Drought levels are indicated using the simulated SIM2 SWIs, which rely on a different set of assumptions⁸. We employed a standardized index, which uses the median of climate models and is freely available on the Safran grid through the Drias-Eau portal.

The standardized SIM2 SWI allows for quick calculation of prospective return periods using a standard normal distribution table. The return period is an intuitive way to express frequency or probability. For example, if a drought of a given intensity is observed five times in 100 years, it is said to have a return period of 20 years. When using climate projections, return periods are calculated over a fixed historical range for which observations are available (1958-2007 for ClimSec). Due to climate change, using return periods in projections characterizes the intensity of a phenomenon, but its interpretation in terms of frequency is only valid for past periods.

two preceding months: a definition based on a sliding period better reflects the slow kinetics of shrink-swell.

⁶The uniform SWI is based on the assumption of identical soil texture and vegetation cover across the entire national territory, specifically grass-type vegetation and predominantly clay soil.

⁷The influence of scenarios (RCP 4.5, RCP 8.5) is not significant until 2050 but becomes substantial in projections up to 2100.

⁸For example, unlike the uniform SWI, the simulated SIM2 SWI takes local topography and vegetation type into account: hence, historical SWIs should be compared with projected values cautiously.

Conversely, when using actual observations for Cat-Nat commissions, the reference period is defined dynamically. For shrink-swell requests between 2018 and 2023, the reference period is the last 50 years (Circular n°INTE1911312C 2019). With the meteorological criterion valid in 2023, a Cat-Nat recognition request in 2084 would use observations from 2055 onwards. The warming trend explains why our method using SWI projections will likely overestimate the number of geotechnical droughts in terms of Cat-Nat recognition. This is not a significant issue, as our goal is to model damage to housing, not to predict claim compensations.

3.1.3 Cat-Nat claims

The database of administrative procedures relating to risks (BD GASPARE) includes all the procedures for recognition of the state of natural disasters, updated within 30 days of their publication in the Journal Officiel. This database, which is heterogeneous in its form, is updated directly by the departmental or national instruction services.

The source used in this study was reprocessed by Ubyrisk Consultants, a private firm specialising in natural hazards. This enhanced database enables us to find out, for each application for recognition of a natural disaster related to shrink-swell since 1982, the outcome of the application (accepted, refused), the start and end dates of the event, and the date of the interministerial decree ruling on the matter.

This dataset, which includes both accepted and rejected applications, allows us to evaluate the quality of our replication exercise for events that occurred between January 1, 2018, and December 31, 2022. With the exception of four municipalities excluded from the sample due to mergers during this period, our findings align closely with those of the interministerial commission. Of the nearly 70,000 quarters assessed by the commission, we identified 584 false positives (0.86% of the sample), where our replication suggested recognition but was ultimately refused, and 410 false negatives (0.60% of the sample). The occurrence of false positives is not particularly concerning, as procedural errors—such as an application for recognition spanning

two calendar years—can lead to a rejection despite meeting the criteria. However, the presence of false negatives warrants further investigation; potential explanations may include variations in the geographical boundaries of municipalities, procedural errors or manipulation. The relatively small discrepancy does not undermine the validity of our results. Nevertheless, it is important to acknowledge the limitations of the BD Gaspar, which stem from its administrative objectives, particularly regarding the imprecision in the classification of hazards. These limitations may impact the reliability and comprehensiveness of the data (Douvinet 2006, Cazaux, Meur-Férec, and Peinturier 2019).

Additionally, it is noteworthy that the meteorological criterion is very frequently met in a municipality without a corresponding request for the recognition of a natural disaster. This highlights that validation of the meteorological criterion at the municipal level does not necessarily imply the occurrence of damage, particularly in areas with few or no individual houses built on clay foundations.

3.2 Real estate data

In order to assess the effect of the exposure to shrink-swell risk and of the related insurance framework on dwellings, we use two sources: the Fidéli file, which exhaustively contains the characteristics of existing dwellings and households, and the DV3F processing, which includes property transactions between 2010 and 2022.

3.2.1 Fideli - Dwellings and their inhabitants

The Fidéli database is a comprehensive, systematically assembled collection of administrative data primarily derived from French fiscal records (Lamarche and Lollivier 2021). It is designed to enhance the statistical understanding of housing and resident demographics in France. The database integrates various fiscal datasets, including property tax files, records of built properties, income tax returns, and income declarations, all linked through fiscal identifiers corresponding to households and properties.

In addition to fiscal data, Fidéli incorporates supplementary contextual information

that enriches its dataset. This includes variables that describe the specific locations where individuals reside, such as geographic coordinates, urban planning identifiers (IRIS codes), and the presence of community housing. Moreover, the dataset offers aggregated income data, including salaries, pensions, and social benefits received by households. These details are available at both the individual and household levels, supported by the Filosofi framework.

As part of this project, we have enhanced Fidéli by integrating data on natural disasters from various risk categories, including droughts, floods, heatwaves, wildfires, and sea level rise. We have specifically chosen the 2017 version of Fidéli to utilize the established link between dwellings and their owners (André and Meslin 2022). This dataset will be made available to researchers.

3.2.2 DV3F - Transactions

The DV3F (Demande de Valeurs Foncières enrichie) database collects comprehensive data on property transactions throughout France since 2010, except in Alsace, Moselle, and Mayotte. This dataset builds upon the foundational DVF (Demande de Valeurs Foncières) file managed by the French General Directorate of Public Finances (DGFIP), which is sourced from notarial deeds and enhanced with cadastral information. DVF provides detailed insights into property transactions, including sale prices and property characteristics. However, its raw format initially posed challenges for conducting advanced economic analyses.

To address these limitations, the DV3F database was developed through a collaborative effort led by the Centre for Studies and Expertise on Risks, Environment, Mobility, and Urban Planning (Cerema). Introduced in 2017, the DV3F database improves upon the original DVF data by structuring the file (1 line per transaction), incorporating geolocation information using parcel geometries from the National Institute of Geographic and Forestry Information (IGN) and adding enriched cadastral data from the BD MAJIC files. By conducting this project at INSEE, we gained

access to all recorded transactions from 2010 to 2022 in the DV3F database.

To accurately assess the impact on prices, it is essential to first filter the database to extract a subset of comparable transactions (Casanova Enault and Peris 2022). In the initial stage of this filtering process, we build on the work of the INSEE PSAR SL team, which created tools to systematically analyse property markets. To create a subset that accurately reflects the local property market, PSAR SL focuses exclusively on residential properties (houses, flats, building plots) purchased by individuals, representing 76% of residential property buyers. This approach is particularly relevant for our study, as we are primarily concerned with household responses to natural disaster insurance.

The PSAR SL team navigated a crucial trade-off: excluding atypical properties, such as those with exceptional size or number of rooms, to avoid skewing price analyses, while retaining a sufficient number of transactions to ensure robustness. Their work led them to include only sales in the secondary market, excluding, among others, estates and expropriations, properties with less than 9 m² or more than 200 m² for flats and 300 m² for houses, multiple properties sold over 1 km apart or in different municipalities, transactions priced at zero or 1€, and properties sold twice on the same day. Following the application of these filters, 48.2% of the transactions were retained for analyzing the secondary real estate market, with many exclusions due to the nature of the properties (e.g., land, professional premises, or outbuildings). Transactions with extreme prices are also filtered out. This process resulted in a dataset of 9.1 million sales, of which we focus on 5.3 million houses, as individual houses are most susceptible to clay swelling damage.

House prices are influenced by external factors such as location, nearby services or nuisances, and neighbourhood demographics. The PSAR SL team further enriched the subset with additional data sources, including geo-referenced variables on a 200m grid and data at the Iris or municipal level (full list of variables in the Appendix 9.3).

4 Empirical Strategy

The empirical strategy employed aims to address two key questions: First, is the risk of clay shrinkage and swelling reflected in house sale prices between 2010 and 2022, and how is it likely to evolve? Second, how critical in real estate markets is the insurance mechanism linked to the recognition - or the lack thereof - of a natural disaster due to clay shrinkage and swelling ?

To answer the first question, the analysis will rely on descriptive and inferential examinations of historical data and future projections. The second question will be approached using a quasi-experimental design.

4.1 Conceptual framework: Hedonic modeling and determinants of housing prices

Housing is a complex good characterized by a network of constituent attributes (size, number of rooms, age, location, energy efficiency) and amenities (neighbourhood, services closeby), which makes single homes hardly substituable (Rosen 1974, Sirmans, Macpherson, and Zietz 2005). On average, the annual housing turnover rate stands at 2.5% (DV3F 2010-2022): comparing average real estate prices across regions or periods is therefore very likely to be insufficient as it conflates price effects with differences in composition. Hedonic pricing is essential for capturing the effect of exogenous shocks on real estate prices as it isolates these attributes. Controlling for observable differences in utility is a necessary albeit non sufficient step for a clear identification of price effects (Casanova Enault and Peris 2022).

The selection of relevant variables is guided by the following trade-off: enhancing the model's accuracy by including as many economically pertinent variables as possible to explain the price, while avoiding overfitting and mitigating the risk of reduced accuracy due to multicollinearity. The variables chosen pertain to the specific characteristics of the house being sold, the neighbourhood and municipality in which it is

located and their surroundings, as well as year and municipality fixed effects. Selection of covariates, based on multicollinearity tests and the computation of variance inflation factors (VIF), is detailed in the Appendix (9.3). The chosen covariates have VIF values ranging from 1 to 3, indicating that multicollinearity is not a significant issue.

A linear model is generally unsuitable because it assumes that the price change resulting from a one-unit increase in a characteristic is constant, regardless of the characteristic's level (Zhang and Leonard 2018). For instance, it presumes that the impact on a property's sale price from an additional square meter of living space is the same whether increasing from 50 to 51 square meters or from 150 to 151 square meters. This assumption overlooks the diminishing marginal utility of additional space, leading to inaccurate estimations. Our baseline hedonic model is given in equation 1, with $X_{i,j,t}^k$ the controls, δ_j municipality fixed effects, λ_t year fixed effects. It achieves an R^2 of 0.75, suggesting that it captures a substantial portion of the variance in house prices without overfitting (see Table 7 in Appendix 9.3).

$$\log p_{i,j,t} = \beta_0 + \sum_{k=1}^K \beta_k X_{i,j,t}^k + \delta_j + \lambda_t + \varepsilon_{i,j,t} \quad (1)$$

4.2 Current and future exposition to shrink swell: a question of salience

We employ a hedonic pricing model to analyze housing sale data, aiming to determine whether the risk of clay swelling is reflected in real estate prices. We first perform a naive OLS regression to evaluate in-pricing of shrink-swell risk. We consider a house as exposed to the clay related subsidence if it is located in an area of medium to high exposure according to the zoning carried out by the BRGM, i.e. $\text{Clay}_i = 1$ in equation 2:

$$\log p_{i,j,t} = \eta_0 + \eta_1 \text{Clay}_i + \sum_{k=1}^K \beta_k X_{i,j,t}^k + \delta_j + \lambda_t + \varepsilon_{i,j,t} \quad (2)$$

Since January 1, 2020, France has required soil studies to assess clay swelling risk when selling buildable plots or existing houses in designated at-risk areas. This mandate, established by the ELAN law of November 23, 2018, and further specified by the decree of July 22, 2019, ensures that the potential impact of this geological risk is transparently communicated through the inclusion of a soil survey in sales agreements. Although soil surveys existed prior to this requirement—such as the soil sensitivity maps produced by BRGM between 1997 and 2010—these were primarily intended for public authorities rather than individual homeowners.

This reglementary shock allows our hedonic pricing model to assess if and how this risk is reflected in property prices, which is what we test in equation 3:

$$\log p_{i,j,t} = \gamma_0 + \gamma_1 \text{Clay}_i + \gamma_2 \text{Clay}_i \times \text{post-ELAN}_t + \sum_{k=1}^K \beta_k X_{i,j,t}^k + \delta_j + \lambda_t + \varepsilon_{i,j,t} \quad (3)$$

To ensure our estimate ($\hat{\gamma}_2$) of the impact of clay swelling risk on property prices is causal, several assumptions are necessary. First, the conditional parallel trends assumption must hold, meaning that conditional on the controls in the hedonic pricing model, property prices in exposed and non-exposed areas would have followed similar trends in the absence of the ELAN law. Second, there should be no other simultaneous interventions that could differentially affect these areas. Third, we assume no anticipation effects : before the Elan law, soil survey results at the plot level were hardly accessible.

This conditional difference-in-difference regression facilitates an assessment of whether property prices have exhibited distinct trends from 2020 onwards between houses built on soil exposed to shrink-swell compared to those not exposed. Nevertheless, it does not offer insights into the dynamics of the risk linked to clay shrinkage and swelling. To address this, we introduce a dynamic concept of shrink-swell exposure by incorporating a trigger factor. In practice, we replicate the recognition process of

a natural disaster resulting from differential subsidence caused by clay shrinkage and swelling, utilizing SWI projections and the return period thresholds (c^{rp}) defined in the Circular n°INTE1911312C 2019:

$$\widehat{\text{Clay}}_{i,j,t} = \text{Clay}_i \times I(\widehat{\text{SWI}}_{i,t} > c^{rp}) \quad (4)$$

In our prospective statistics, the focus shifts from merely identifying real estate transactions to locating households. Within this framework, a household is considered at risk if its primary residence is located in an area with weak to high exposure, as delineated by the BRGM map, and if the Safran grid encompassing the residence experiences superficial soil drought with a return period of at least 25 years. We focus on two key indicators: the number of dwellings potentially at risk of damage over various time horizons (reference period, 2021-2050, 2041-2070, 2071-2100), and the proportion of the population residing in a municipality that would have been officially recognised as experiencing a natural disaster under the legislation effective until 2023.

The rationale is as follows: the perceived importance of a risk can be shaped by the damage directly experienced by households, by the experiences of those in their vicinity (either directly or through intermediaries such as media coverage), or by regulations imposed by public authorities (e.g., the Elan law, which mandates transparency in risk information). However, several caveats must be considered: this scenario does not account for migratory patterns or changes in building location and quality. Therefore, the results should be interpreted as illustrative of what might occur in the absence of coordinated efforts to adapt to increasing risks, rather than being viewed as predictive in nature.

We also examine the socioeconomic characteristics of individuals affected by natural hazards over time, with a particular emphasis on their standard of living. Presently, the average cost of repairs is estimated at 16,500 euros, equivalent to 72% of the

median annual standard of living (Insee, 2021). As projections suggest that this cost will rise (Cour des comptes 2022, CCR 2023), the issue of the distributive impact of natural risks becomes increasingly significant. The existing public-private insurance framework, which distributes these costs across all households rather than solely burdening those whose properties are damaged, is central to discussions of climate justice, particularly in the context of shrink-swell phenomena.

4.3 Assessing the role of public reinsurance: the impact of natural disaster recognition on real estate prices

4.3.1 Disentangling Insurance from Damage

Given a well-defined identification strategy, we anticipate that the effect of shrink-swell risk cannot be positive, as it is typically perceived as a disamenity that unambiguously reduces utility. The extent to which it is negative is likely contingent on how homeowners are informed about and confident in the insurance mechanisms. In contrast, the impact of the Cat-Nat insurance mechanism is *ex ante* ambiguous. An initial approach involves examining the correlation between the cumulative number of Cat-Nat recognitions at the municipal level ($R_{j,t}$) and property prices, while controlling for dwelling characteristics, environmental factors, and the presence of clay beneath foundations. We test this specification with equation 5:

$$\log p_{i,j,t} = \mu_0 + \mu_1 R_{j,t} + \mu_2 \text{Clay}_i + \sum_{k=1}^K \beta_k X_{i,j,t}^k + \delta_j + \lambda_t + \varepsilon_{i,j,t} \quad (5)$$

However, merely observing the dominance of a particular signal does not constitute definitive evidence. In a scenario with perfect information, and when controlling for actual shrink-swell damage, we expect the effect of insurance to be positive, as it provides compensation for incurred damages. Conversely, in a setting characterized by imperfect information, the net effect of insurance is ambiguous *ex ante* because the mechanism may heighten risk awareness, thereby potentially depressing prices.

Furthermore, without plot-level data on damages, Cat-Nat recognition might reflect the severity of damage within a municipality, exerting downward pressure on prices.

The gradual progression of subsidence phenomena makes it challenging to accurately date the damage at the residential plot level. Instead of relying on the declared start or end date of an event at the municipal level, we will focus on the publication date of the Cat-Nat recognition order to determine whether the various requests are accepted. Once the interministerial order is published in the *Journal Officiel*, policyholders have 30 days to submit their claims to their insurance companies. The insurance companies then have three months from the publication date to process and pay compensation. Thus, any positive effect associated with insurance coverage is likely to become apparent in property prices only after the decree’s publication.

4.3.2 A quasi-experimental design

Disentangling the effects discussed above is challenging, but the regulatory framework provides a clearer perspective. The binary nature of the meteorological criterion, with a 25-year soil drought return period cutoff between 2018 and 2023, presents an opportunity for researchers. Practically, this criterion requires the Soil Wetness Index (SWI) for a given month to be the lowest or second lowest over a 50-year period. The extent of litigation and the subsequent revision of the criterion to a 10-year return period suggest that damage occurs on clay soils even during less severe droughts. To put it bluntly, the lack of precision in the regulatory criterion allows us to evaluate the impact of being denied insurance coverage despite the occurrence of damage.

First, we compare two groups of municipalities that have applied for Cat-Nat recognition: those whose assessed SWI is the lowest or second lowest in the past 50 years are treated, while those with the third or fourth lowest SWI serve as the control group. The underlying logic is that of a regression discontinuity design: both groups have experienced damage⁹, and in the absence of detailed damage data for each municipality,

⁹The control group has 17 and 13-year return periods, which would, under the new set of rules, suffice to obtain recognition.

our identification hypothesis assumes that the damage extent is comparable across groups. Under this assumption, we can isolate the effect of the insurance mechanism itself, or conversely, the effect of being denied insurance coverage. Equation 6, our primary Difference-in-difference specification, is run on the selected groups, as well as on a subset of houses built over clay soil: obtained estimates are therefore LATEs¹⁰.

$$\log p_{i,j,t} = \alpha_0 + \alpha_1 \text{Treatment}_j \times \text{Years to treatment}_{j,t} + \sum_{k=1}^K \beta_k X_{i,j,t}^k + \delta_j + \lambda_t + \varepsilon_{i,j,t} \quad (6)$$

Our replication of the commission's work did not achieve a 100% success rate : Figure 2b highlights the fuzzy nature of treatment, eventhough the SWI rankings are continuous (Figure 2a). Due to concerns about potential manipulation by municipalities of the commission's decision, which could bias our findings, we define our treatment dummy solely with the SWI rankings, rather than true recognition. Conceptually, we instrument the recognitions and rejections and conduct reduced-form regressions.¹¹

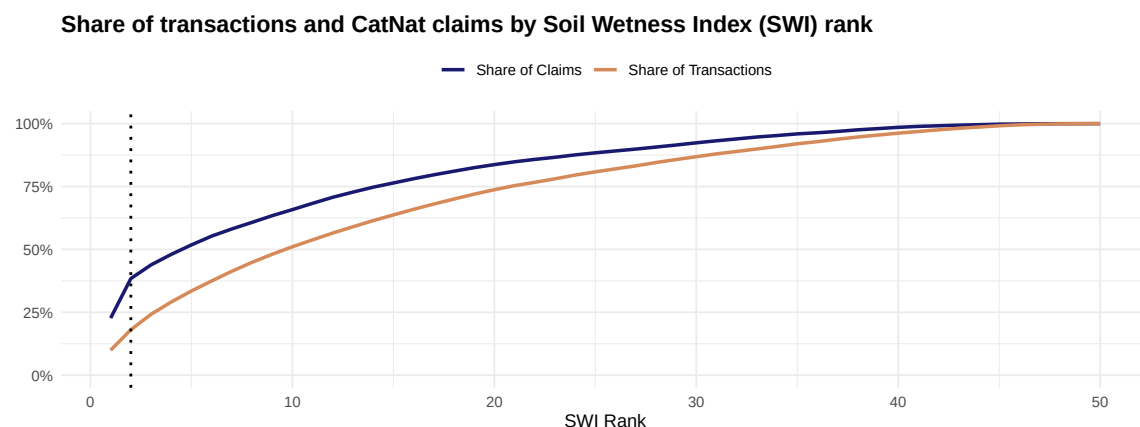
Second, we aim to assess the differential impact of clay-related damage within a given treatment group (i.e. municipalities in J_{treated} or $J_{\text{untreated}}$). Specifically, among municipalities recognised as natural disaster areas or denied recognition, we evaluate the effect at the property level of being situated on clay-rich patches. Equation 7 provides insight into the underlying mechanisms: irrespective of the insurance response, how do the damaged houses perform in the market?

$$\log p_{i,j,t} = \theta_0 + \theta_1 \text{Clay}_i \times \text{Years to treatment}_{j,t} + \sum_{k=1}^K \beta_k X_{i,j,t}^k + \delta_j + \lambda_t + \varepsilon_{i,j,t} \quad (7)$$

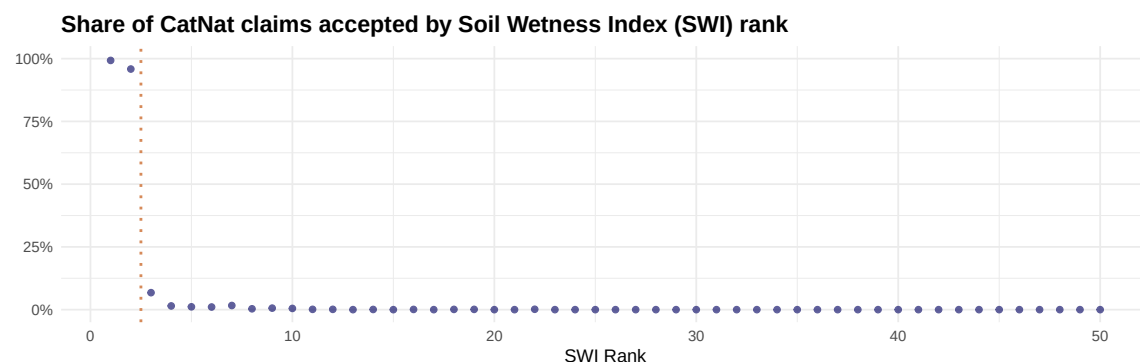
for $j \in J_{\text{treated}}$ or $J_{\text{untreated}}$

¹⁰In this equation, there is no Treatment dummy outside of the interaction term of interest: since treatment is administered at the municipality level, including it alongside municipality fixed-effects would lead to perfect multicollinearity.

¹¹In practice, our results remain consistent when we include only recognised municipalities in the treatment group and non-recognised municipalities in the control group.



(a) Minimal SWI is a continuous variable



Note: The SWI assessed is the lowest across all Safran grids intersecting with municipality area for the period over which recognition of the state of natural disaster is requested. Rank 1 and 2 correspond to a return period over 25, and should be granted recognition. Data: SWI, Météo France; BD Catnat. Replication made by the author.

(b) The 25-year return period threshold is not completely sharp

Figure 2: Fuzzy treatment

Given that natural disasters are recognised quarterly, and sometimes repeatedly, the challenge of selecting an appropriate time frame for analysis becomes particularly significant. The possibility for treatment to be repeated and our aim to compare municipalities that may have experienced damage result in varying groups from one season to the next. This variability makes it especially challenging to implement a staggered Difference-in-Differences (DiD) approach or an event study on the full data set within a clear inferential framework.

Our primary specification is therefore limited to the recognitions that occurred in

2019, for two key reasons. The change in meteorological criterion in 2019 (Circular n°INTE1911312C 2019) led to a bottleneck, resulting in the assessment of many requests for older phenomena within that year. Consequently, between 2010 and 2022, 815,000 house sales occurred within our target groups. These groups include all municipalities whose applications for natural disaster recognition were processed in 2019, with their minimum SWI return periods either satisfying the meteorological criterion (ranks 1 and 2) or being proximate to the threshold (ranks 3 and 4). This methodological approach enables us to maximize statistical power while maintaining a unique threshold year. Beyond this econometric rationale, the recognitions made in 2019 are of particular interest due to the significant increase in litigation related to Cat-Nat decrees that followed (Cour des comptes 2022).

In a complementary analysis, for each of the 15 quarters studied (covering all quarters from 2019 to 2021 and the first three quarters of 2022), we select municipalities assessed under the same decree and form our groups accordingly. We then adjust the timing of transactions by expressing it as the number of years between the transaction and the treatment. Finally, we aggregate these 15 datasets to enhance the statistical power of our analysis. This approach leads to some transactions appearing multiple times across the datasets, a limitation that will be discussed in the subsequent section.

5 Results

5.1 Shrink-swell: a salient risk whose significance is set to soar

5.1.1 Medium to high exposure to clay subsidence now leads to a 2% discount on house sale prices

Between 2010 and 2022, the shrink-swell risk, as delineated by the BRGM's exposure categories, exhibits a positive correlation with property prices (Table 1, OLS without controls). However, there is no theoretical basis to suggest that clay-rich soils should lead to an overvaluation of the houses constructed on them, indicating the presence of

omitted variable bias. Upon applying our hedonic model, the coefficient reverses its sign: the current risk is associated with a 0.7% decrease in sales prices over the period from 2010 to 2022 (Table 1, OLS with controls). For the median sale price, which stands at €251,000 in 2021-2022, this corresponds to a reduction of approximately €1,750.

This coefficient is challenging to interpret in economic terms, as the relevant data was not readily accessible prior to 2020. Consequently, our hypothesis posits that this effect is predominantly influenced by transactions occurring after 2020. Our conditional difference-in-differences analysis confirms the absence of significant effect before the ELAN law (Table 1, DiD). After 2020, shrink-swell risk — represented by the medium to high-risk categories, which account for 93% of damages (CCR 2024) — results in a 1.7% decrease in prices, equivalent to a reduction of 4,250 euros on the median house price. For comparison, a meta-analysis of price discounts associated with flood zones suggests an average price decrease of 4.4% (Assen 2024). This aligns with the intuition that the risk associated with clay shrinkage and swelling is still underrecognised, yet its significance has increased in recent years.

This coefficient can only be interpreted causally if the conditional parallel trends hypothesis is valid. Figure 3 suggests that this is only partially the case: the effect of the exposure category is effectively zero between 2012 and 2018, but the effect is significant in 2010 and 2011.

There are several potential explanations for this unexpected effect. The first is to seriously consider the possibility of a temporary recognition of the risk in 2010-2011. This could be attributed to a confluence of factors, including the completion of the initial wave of soil surveys, changes in the recognition procedure in 2010, and the occurrence of significant droughts in 2009 and 2011. To illustrate, nearly 1,800 municipalities applied for natural disaster recognition in 2009, and this number surged to 5,300 in 2011, compared to fewer than 600 applications in 2007-2008 and 2013-2014.

Table 1: In-pricing of shrink-swell risk in French house prices (2010-2022): OLS

Dependent Variable: Model:	(No controls)	House prices (log) (OLS with controls)	(DiD)
<i>Variables</i>			
Exposed to shrink-swell	0.136*** (0.013)	-0.007*** (0.002)	-0.002 (0.002)
Exposed $\times$ Post ELAN law			-0.017*** (0.002)
House characteristics	No	Yes	Yes
Neighbourhood characteristics	No	Yes	Yes
Local amenities	No	Yes	Yes
<i>Fixed-effects</i>			
Year of transaction	Yes	Yes	Yes
Municipality	No	Yes	Yes
<i>Fit statistics</i>			
Observations	5,251,626	5,232,345	5,232,345
R ²	0.019	0.755	0.755
Within R ²	0.009	0.495	0.495

Clustered (Municipality) standard-errors in parentheses

Significance Codes: ***: 0.01, **: 0.05, *: 0.1

Source: Metropolitan France house sales (2010-2022), DV3F; enriched by Insee PSAR-SL; Shrink swell exposure maps (BRGM)

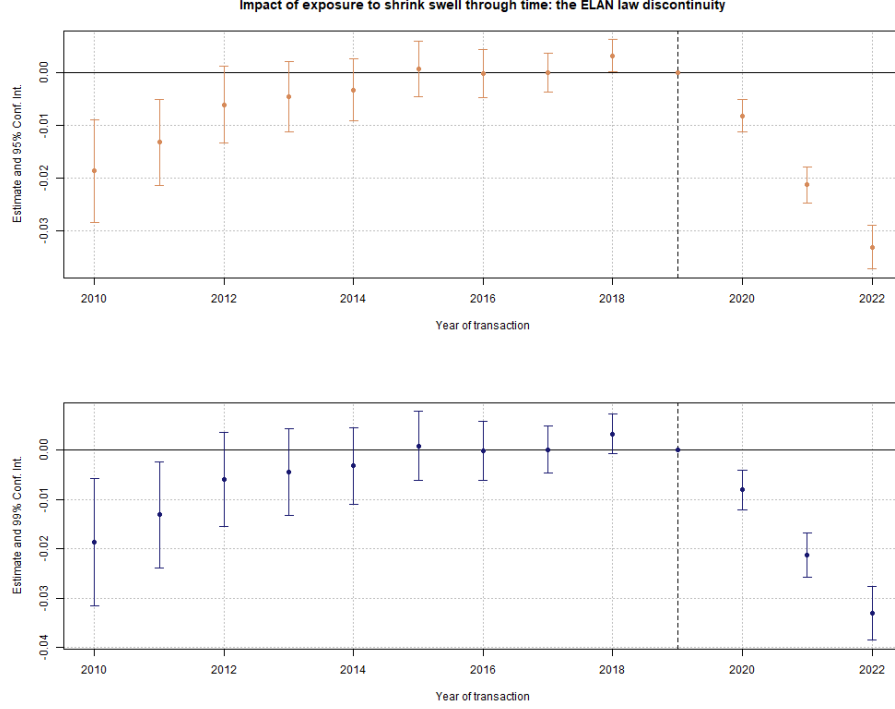


Figure 3: Assessing pre-trends at the 95% and the 99% confidence interval

The second set of explanations, which the author favors, involves the suspicion of a spurious correlation that the hedonic model's controls, calibrated for a study conducted in the 2020s, fail to account for. While the characteristics of the homes are reflective of their state at the time of the transaction, the characteristics of the neighbourhoods and surrounding areas are indicative of conditions at the end of the study period. For instance, the implementation of Seveso sites may have changed, priority urban neighbourhoods only came into existence after 2015, and transportation stations may have been opened or closed during this time.

In conclusion, we find strong suggestive evidence supporting a moderate negative impact of shrink-swell risk on real estate pricing. However, it is important to acknowledge that the possibility of omitted variable bias in our results cannot be excluded.

5.1.2 The proportion of households directly or indirectly exposed to clay shrinkage and swelling is expected to rise sharply

Out of the 19.3 million houses in metropolitan France (Fideli 2017), 15.8 million are situated on clay-rich soil. Among these, using our dynamix proxy for damage $\widehat{\text{Clay}}_{i,j,t}$ defined in equation 4, we estimate that 1.1 million homes will face at least one exceptional drought likely to trigger shrink-swell between 2021 and 2050, a figure close to the modeled reference period (1.6 million). This number is projected to rise to 3.6 million between 2041 and 2070, and to 14 million between 2071 and 2100, according to our computations based on the IPCC RCP 8.5 scenario. In otherwords, three fourths of single-homes and almost all houses located on clay soils are likely to face damage. On the intensive margin, the median number of exceptional droughts faced remains at 0 in 2021-2050 and 2041-2070, but increases to 4.3 between 2071 and 2100.

The observed number of houses compensated each year is not made public, nor is the total number of houses facing damage. However, with the total amount of insurance relief fluctuating in recent years between 300 million euros (2016) and 3.5 billion euros (2022, CCR 2024), and an average repair cost of approximately 16,000 euros (Cour des comptes 2022), it can be estimated that between 18,000 and 220,000 houses are damaged annually. In our modeled reference period, using the same figures, we estimate a yearly average of 53,000 houses likely to be damaged. If our ratios hold, one expects around 466,000 houses likely to be damaged every year towards the end of the century, or on average, a drought twice as extreme as the 2022 historical drought, every year.

These figures should not be taken at face value, for there are numerous sources of uncertainty, such as the proportion of houses likely to be damaged that actually sustain damage, the effectiveness of efforts to mitigate climate change, potential migratory responses, and improvements in house foundation designs. Nevertheless, this provides

insight into the future dynamics: shrink-swell damage is set to soar. Moreover, according to the CCR’s projections for 2050, the majority of the rise in claim costs is attributed not to the intensification of climatic hazards, but rather to the increased value of insured assets (CCR 2023). The geolocated results presented in the Appendix 9.4.1 illustrate the magnitude of changes according to the IPCC’s 4.5 and 8.5 scenarios. Across all scenarios, we observe both an expansion of risk into regions historically less prone to drought (particularly in the northern half of the country) and an intensification of the risk.

This amplification dynamic has two significant implications for our study. First, it suggests that the salience of this risk is poised for substantial growth. Beyond the direct impact, which, as we have observed, is expected to change by an order of magnitude, the indirect impact is also likely to intensify, i.e. on neighbours or public infrastructure. Figure 4 illustrates the 10-year average of the share of population living in municipalities where both geological and meteorological criteria would be met at least once in the year. What was once a marginal phenomenon is projected to become a common occurrence. However, it is important to note a limitation: return periods are calculated based on historical data rather than dynamic periods. Consequently, this result should be interpreted as an indication of potential future damage expansion rather than as a direct indicator of insurance relief.

Lastly, this growing risk posed by climate change is not limited to a particular segment of the population. As illustrated by graphs in Appendix 9.4.2, by 2100, nearly 40% of households will be vulnerable to damage¹². Although the first two income deciles are less exposed due to higher rates of apartment living, this threat shows minimal variation across different living standards. In this context, the effectiveness of the insurance mechanism becomes critical, not only because the potential cost of repairs could exceed the financial capacity of many households, but also because this risk

¹²Results shown follow the IPCC RCP 8.5 carbon scenario. RCP 4.5 results are available on the Github repository or upon request. (<https://github.com/Sacha-N/Shrink-swell-public-repo.git>)

endangers the long-term viability of the insurance system.

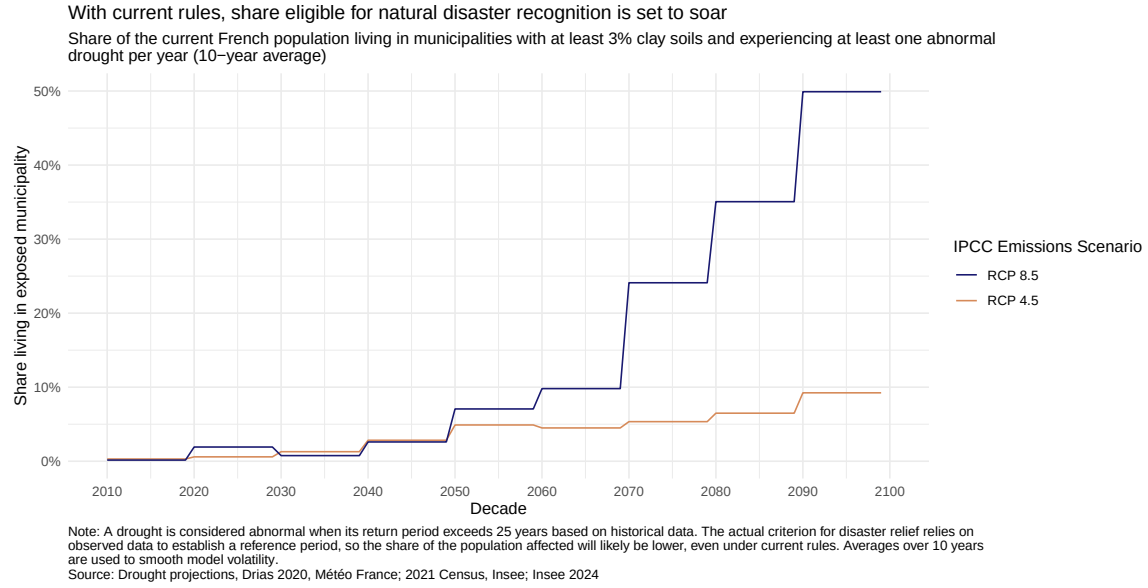


Figure 4: Projected share living in exposed municipalities through time

5.2 Natural disaster insurance: relief or signal?

5.2.1 Are past natural disasters priced-in ?

To better understand the potential implications of removing the risk of clay swelling from the Cat-Nat system, it is essential to first comprehend the significance of its current operation. The problem can be intuitively framed as determining whether the recognition of natural disasters within each municipality is reflected in the housing prices in those areas. To explore this, Table 2 demonstrates that there is no strong correlation between housing prices and the cumulative number of disaster recognitions. After accounting for the standard controls in our hedonic model, neither the number of recognitions since the significant drought of 2003 (Table 2, column 1) nor those following the adoption of current criteria in 2018 (Table 2, column 3) have a significant impact on housing prices, either directly or through interaction with average or high exposure to the shrink-swell phenomenon.

Table 2: In-pricing of shrink-swell insurance in French house prices (2010-2022): OLS

Dependent Variable: Model:	House prices (log)		
	(1)	(2)	(3)
<i>Variables</i>			
Exposed to shrink-swell	-0.009*** (0.002)	0.012*** (0.004)	-0.007*** (0.002)
Number of past Cat-Nat recognitions (2003)	0.002 (0.003)	0.0003 (0.004)	
Past Cat-Nat recognitions (2003) $\times$ Exposed	0.002 (0.002)	0.003 (0.004)	
Past Cat-Nat recognitions (2018)			-0.0006 (0.001)
Past Cat-Nat recognitions (2018) $\times$ Exposed			-0.0002 (0.001)
House characteristics	Yes	Yes	Yes
Neighbourhood characteristics	Yes	Yes	Yes
Local amenities	Yes	Yes	Yes
<i>Fixed-effects</i>			
Municipality	Yes	No	Yes
Year of transaction	Yes	Yes	Yes
Department	No	Yes	No
<i>Fit statistics</i>			
Observations	5,232,345	5,231,900	2,373,719
R ²	0.755	0.709	0.769
Within R ²	0.495	0.576	0.500

Clustered (Municipality) standard-errors in parentheses

Significance Codes: ***: 0.01, **: 0.05, *: 0.1

Source: Metropolitan France house sales (2010-2022), DV3F; enriched by Insee PSAR-SL;

Shrink swell exposure maps (BRGM); BD Cat-Nat

Several potential explanations could account for this lack of correlation. It is conceivable to hypothesize that homeowners expect full insurance coverage in the event of damage. However, this assumption relies on the presence of perfect information, a recognition process free of false negatives, and the absence of any non-monetary valuation of dwellings.

One other possibility is that the temporal variation in the number of recognitions per municipality is insufficient, leading the fixed effects at the municipality level to capture the entire effect. However, when applying fixed effects at the *département* level (Table 2, column 2), the coefficient remains insignificant. Moreover, excluding the municipality-level fixed effects reveals a positive effect of shrink swell exposure on housing prices, strongly indicating the presence of omitted variable bias. For example, municipalities with a high proportion of clay soils might engage in different agricultural practices, which could influence the local economy and, in turn, the real estate market.

Two additional challenges hinder the effectiveness of this strategy. First, this methodology disproportionately emphasizes municipalities that have experienced the largest increase in recognitions, whose property markets may respond differently, potentially due to a habituation effect. More critically, it fails to distinguish between the damage effect and the insurance effect.

5.2.2 The impact of 2018-2019 droughts

In light of these considerations, our analysis centers on a set of recognitions, specifically those that occurred in 2019. Table 3 presents the results of our primary specification : a difference-in-difference with controls corresponding to equation 6. In the first column, we observe that, with the inclusion of controls from our hedonic model, the recognition of a natural disaster has no impact on prices prior to the interministerial decree, supporting the credibility of the conditional parallel trends hypothesis. Following the decree, there is a progressively positive effect. Two years after the treat-

ment, houses in the treated group sell for 1% more, and this increase rises to 2.5% three years later. This suggests a significant insurance effect: assuming the damage is comparable between the two groups, which corresponds to one of our identification hypotheses, we demonstrate that being denied insurance has a substantial impact on prices, leading to a reduction in selling prices by approximately 6,000 euros on the median sale price.

To strengthen the credibility of this interpretation, the third column replicates the analysis on the subgroup of houses categorized as having medium or high exposure to clay shrinkage and swelling. Our results remain largely consistent; however, it is worth noting that the three-year effect is slightly more pronounced in this subgroup, which may lend additional support to our interpretation. A graphical representation of the effects of Cat-Nat recognition for each year from treatment is provided in the Appendix 9.5.1. For the specifications corresponding to columns 1 and 2 of Table 3, graphs are provided with standard errors clustered at the municipality level (Figure 21).

This interpretation, while credible, does not exhaust all possibilities. The table 4 advises caution in interpreting our results. Within the treated group (column 1), that is, the municipalities recognised where residents were able to claim compensation from their insurance companies, the pricing of shrink-swell exposure is less clear than in the general population. While it was at -1.7% from 2020 onwards (Table 1), the effect appears to be negligible in 2019-2020 and even turns positive in 2022. In the untreated group (column 2), that is, among homes that were potentially damaged but without compensation, the pricing remains negative up to two years before the recognition order, after which it becomes less distinct.

Table 3: In-pricing of shrink-swell insurance in French house prices (2010-2022) - Municipalities requesting Cat-Nat recognition in 2019

Dependent Variable: Model:	House prices (log)	
	(2019 <i>Journal Officiel</i>)	(Only exposed houses)
<i>Variables</i>		
Treatment $\times$ Years to treatment = -5	0.004 (0.018)	0.010 (0.021)
Treatment $\times$ Years to treatment = -4	0.008 (0.015)	0.013 (0.018)
Treatment $\times$ Years to treatment = -3	0.008 (0.010)	0.012 (0.011)
Treatment $\times$ Years to treatment = -2	0.006 (0.004)	0.006 (0.005)
Treatment) $\times$ Years to treatment = 0	-0.002 (0.002)	-0.003 (0.003)
Treatment $\times$ Years to treatment = 1	0.0005 (0.004)	-0.0005 (0.003)
Treatment $\times$ Years to treatment = 2	0.010** (0.004)	0.009** (0.004)
Treatment $\times$ Years to treatment = 3	0.025*** (0.005)	0.028*** (0.005)
<i>Controls</i>		
House characteristics	Yes	Yes
Neighbourhood characteristics	Yes	Yes
Local amenities	Yes	Yes
<i>Fixed-effects</i>		
Municipality	Yes	Yes
Year of transaction	Yes	Yes
<i>Fit statistics</i>		
Observations	814,466	648,215
R ²	0.786	0.790
Within R ²	0.492	0.492

Clustered (Municipality) standard-errors in parentheses

Significance Codes: ***: 0.01, **: 0.05, *: 0.1

Source: Metropolitan France house sales (2010-2022), DV3F; enriched by Insee PSAR-SL; Shrink swell exposure maps (BRGM); Soil Wetness Index (Météo France)

Table 4: Within-group pricing in French house prices (2010-2022) of shrink-swell risk
- Municipalities requesting Cat-Nat recognition in 2019

Dependent Variable: Model:	House prices (log)	
	(Among treated)	(Among untreated)
<i>Variables</i>		
Exposed $\times$ Years to treatment = -5	-0.016** (0.006)	-0.067** (0.028)
Exposed $\times$ Years to treatment = -4	-0.009 (0.006)	-0.053** (0.021)
Exposed $\times$ Years to treatment = -3	-0.001 (0.005)	-0.025** (0.012)
Exposed $\times$ Years to treatment = -2	0.004 (0.005)	0.003 (0.011)
Exposed $\times$ Years to treatment = -1	0.006 (0.004)	0.011 (0.010)
Exposed $\times$ Years to treatment = 0	0.002 (0.004)	0.005 (0.010)
Exposed $\times$ Years to treatment = 1	0.0003 (0.005)	-0.003 (0.011)
Exposed $\times$ Years to treatment = 2	0.005 (0.006)	-0.009 (0.012)
Exposed $\times$ Years to treatment = 3	0.021*** (0.007)	0.003 (0.014)
<i>Controls</i>		
House characteristics	Yes	Yes
Neighbourhood characteristics	Yes	Yes
Local amenities	Yes	Yes
<i>Fixed-effects</i>		
Municipality	Yes	Yes
Year of transaction	Yes	Yes
<i>Fit statistics</i>		
Observations	695,620	118,846
R ²	0.790	0.767
Within R ²	0.491	0.507

Clustered (Municipality) standard-errors in parentheses

Significance Codes: ***: 0.01, **: 0.05, *: 0.1

Source: Metropolitan France house sales (2010-2022), DV3F; enriched by Insee PSAR-SL; Shrink swell exposure maps (BRGM); Soil Wetness Index (Météo France)

To explain these seemingly surprising results, several key points need to be addressed. First, as illustrated in Figure 5, the recognition of natural disaster situations is a lengthy administrative process, with the median time between the end of a declared drought and the issuance of the interministerial decree being 210 days. Unlike the evaluation of the effect of a recognition order, where the timing is clear, it is very difficult to precisely date the damages without specific insurance data. It is plausible that damages may manifest two or three years before the ministerial order, in a significant number of municipalities.

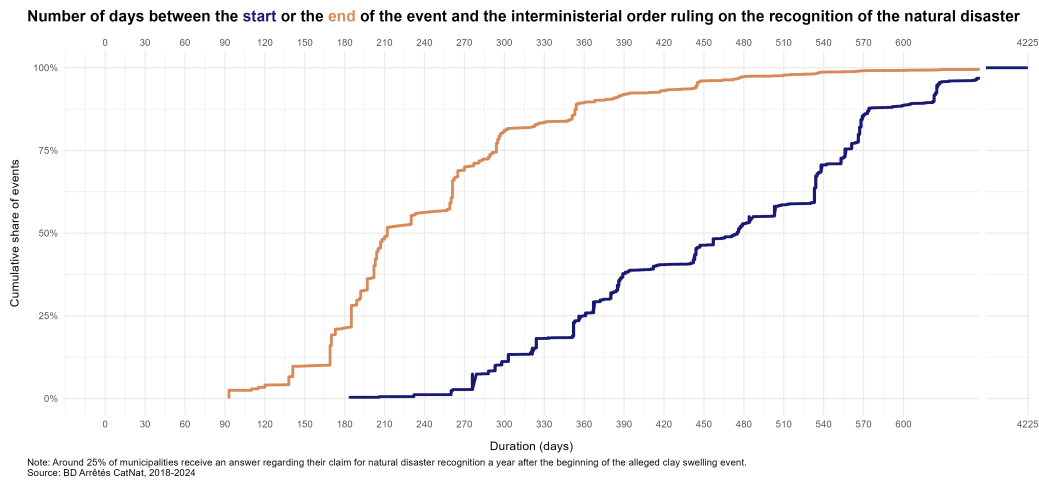


Figure 5: A lengthy administrative process

Next, attention must be paid to the mechanisms by which damages are transmitted into prices. Unlike floods, where the damages can be superficial and concentrated on movable property, the shrinking and swelling of clay can render a house uninhabitable, necessitating extensive and prolonged repairs, which may be undertaken by either the seller or the buyer. Since the enactment of the ELAN law, foundation repairs are often accompanied by preventive measures against subsidence (Cour des comptes 2022), leading to the possibility of a rise in value of certain previously damaged homes. This could explain the positive effects of shrink swell exposure observed at the end of the period in each group. To definitively determine the nature of these effects, precise data on damages and compensations would be required.

Finally, it is crucial to recognise that these subgroup analyses are based on relatively small sample sizes, with fewer than 500 municipalities in the control group. As demonstrated by the summary statistics presented in the Appendix 9.6.1, our groups are not perfectly balanced (see Table 9 and Figure 22): while the hedonic model allows us to control for observable characteristics, we do not have access to all the determinants of price.

5.2.3 Complementary analysis using stacked data

To address these two limitations, a complementary analysis was conducted on all decrees issued between 2019 and 2022. As highlighted with Table 10 and Figure 23 in the Appendix 9.6.2, while still statistically different, the groups are much more alike when one considers transactions from all municipalities in a decree between 2019 and 2022.

Table 5 highlights results that are consistent with those obtained for the year 2019. The within-group analysis also yields comparable trends, as seen in Table 8 shown in the Appendix 9.5.2. However, this method is not without its shortcomings. First, it yields more weight to observations from municipalities which were repeatedly treated, for their transactions appear multiple times in the dataset. As highlighted by the positive slope in Figure 6, municipalities are repeatedly treated. Nevertheless, the parallel trends before treatment suggest that the main source of divergence between both groups is the recognition order assessed.

Perhaps more crucially, by examining the effects of treatments applied at different times simultaneously, it opens the door to the critique posed by Goodman-Bacon 2021. Consequently, these results are valid only if one is willing to assume the homogeneity of recognition effects over time. Given the short timeframe (2019-2022), this assumption might be plausible, but it also requires assuming uniformity across the various groups of municipalities examined. This latter assumption is particularly strong, as the treatment groups are relatively small in many instances, making it

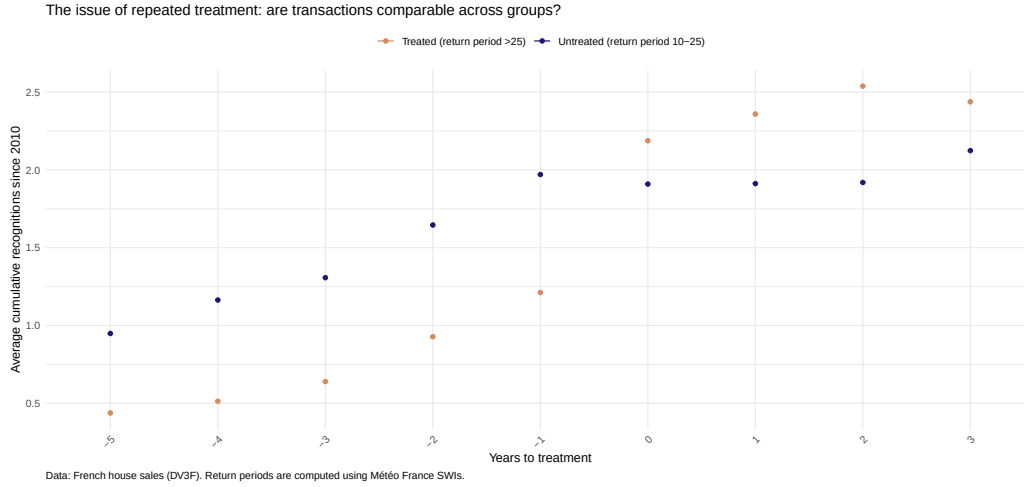


Figure 6: Average cumulative recognitions per group - transactions from all municipalities in 2019-2022 Cat-Nat assessments

plausible that significant differences in observable and unobservable characteristics, which may interact with the treatment, exist.

6 Discussion and mechanisms

6.1 On salience and transmission channels

Since the influential work by Rosen 1974, hedonic pricing models for property valuation have traditionally operated under the assumption of perfect information. Within this framework, households are assumed to select homes that maximize their utility, considering each attribute of the property. Consequently, households would theoretically be willing to pay a premium to avoid natural hazards, unless insurance coverage negates the impact of these risks on property prices. Under these conditions, it is generally anticipated that property prices will maintain their long-term trajectory following a natural disaster, with any price decline being temporary and associated primarily with repair costs (Gwilym Pryce and Galster 2011).

Our findings contribute to a growing body of literature that challenges the adequacy of this theoretical framework, aligning more closely with models that incorporate imperfect information, such as the one developed by Gwilym Pryce and Galster 2011.

Table 5: In-pricing of shrink-swell insurance in French house prices (2010-2022) - Municipalities requesting Cat-Nat recognition between 2019 and 2022

Dependent Variable: Model:	House prices (log)	
	(In municipalities requesting)	(Only exposed houses)
<i>Variables</i>		
Treatment $\times$ Years to treatment = -5	0.0004 (0.006)	-0.0010 (0.006)
Treatment $\times$ Years to treatment = -4	0.0003 (0.004)	-0.001 (0.004)
Treatment $\times$ Years to treatment = -3	-0.0008 (0.002)	-0.002 (0.002)
Treatment $\times$ Years to treatment = -2	-0.0008 (0.0010)	-0.002 (0.001)
Treatment $\times$ Years to treatment = 0	-0.002 (0.001)	-0.002* (0.001)
Treatment $\times$ Years to treatment = 1	0.0004 (0.002)	0.001 (0.002)
Treatment $\times$ Years to treatment = 2	0.007*** (0.002)	0.006*** (0.002)
Treatment $\times$ Years to treatment = 3	0.016*** (0.003)	0.015*** (0.003)
<i>Controls</i>		
House characteristics	Yes	Yes
Neighbourhood characteristics	Yes	Yes
Local amenities	Yes	Yes
<i>Fixed-effects</i>		
Municipality	Yes	Yes
Year of transaction	Yes	Yes
<i>Fit statistics</i>		
Observations	2,367,419	1,919,091
R ²	0.78374	0.78463
Within R ²	0.50412	0.50402

Clustered (Municipality) standard-errors in parentheses

Significance Codes: ***: 0.01, **: 0.05, *: 0.1

Source: Metropolitan France house sales (2010-2022), DV3F; enriched by Insee PSAR-SL; Shrink swell exposure maps (BRGM); Soil Wetness Index (Météo France)

By relaxing the assumption of perfect information and integrating insights from the sociology of risk—particularly regarding the role of institutions in either concealing or disclosing risks—as well as from behavioral economics on how individuals respond to risk (as discussed by Tversky and Kahneman 1973), Gwilym Pryce and Galster 2011 advances a model predicated on the myopic and amnesiac evaluation of property values.

In practical terms, this model offers a conceptual framework where information regarding both anticipated future events and past occurrences is substantially discounted, leading to a divergence between perceived and actual risks. According to Pryce’s model, the effects of myopia and amnesia diminish as natural disasters become more frequent and as information becomes more transparent. This aligns with our projection of a convergence between observed property prices and shrink-swell-adjusted prices in the forthcoming decades.

Drawing on the works of G. A. Tobin and B. E. Montz 1994 and G. Tobin and B. Montz 2007, one may expect this convergence process to be non-linear and to vary locally, driven by an idiosyncratic response to specific Cat-Nat events and their recurrence. In extreme cases, where disasters are infrequent and not repeated, property prices may initially decline but recover after a transitional period. Conversely, in areas experiencing repeated damages, the local market may fail to recover, potentially leading to a downward price spiral. Within this theoretical framework, we would anticipate an intermediate process for the majority of municipalities. However, by 2100, the projected increase in drought frequency could undermine the market’s capacity to recover from each shock, thereby reinforcing the necessity for increased investment in preventive measures and advocating against rebuilding to identical conditions (Cour des comptes 2022).

6.2 Spatial autocorrelation

Any econometric analysis of a spatially localized phenomenon must contend with the issue of spatial autocorrelation, which refers to the dependence between observations in close proximity (Getis 2008). In the context of analyzing property transactions, this typically manifests as positive autocorrelation: property prices of nearby homes tend to exhibit greater similarity. As noted by INSEE’s PSAR-SL, this phenomenon can be attributed to two primary mechanisms: first, homeowners are influenced by the characteristics of the local housing market, and second, when setting a selling price, buyers often reference the prices of properties that have been sold in the vicinity.

The presence of spatial autocorrelation in the residuals can lead to inefficiencies in the OLS estimators, and may even introduce bias if the model fails to account for an explanatory variable that is spatially correlated with the variable of interest (e.g., the amount of sunlight received by each house in our study).

In our analysis, we have opted to cluster the standard errors by municipality. This decision is grounded in the fact that treatment is administered at the municipality level, and we aim for our clustering to reflect the spatial structure of the data that underpins the autocorrelation. By doing so, we relax the assumption of no correlation, but only at the level of the municipality. Furthermore, incorporating fixed effects at the municipality level enables us to capture unobserved determinants that are stable over time and that operate at the municipal level.

One limitation of this approach is the potential for correlations at a more granular scale, such as within neighborhoods, or at the borders between municipalities. An alternative method for addressing spatial autocorrelation would be to incorporate a distance matrix between each transaction. This approach is employed by Zhang and Leonard 2018, where the error term is modeled using a spatial autoregressive process based on a distance matrix to account for spatial autocorrelation in house prices.

6.3 Other limitations and potential improvements

There are several avenues through which this research could be enhanced to increase the robustness of the results. Firstly, the inability to access plot-level damage data, due to the absence of a timely agreement with the Caisse Centrale de Réassurance (CCR), represents a significant limitation. Such data would have allowed us to directly verify our identification hypothesis regarding the similarity in damage extent between the treated and untreated groups. In lieu of this, data on repairs or specific building permits could serve as a more accurate proxy for assessing damage.

Moreover, a longer historical perspective on recent changes, such as the implementation of the Elan law in 2020, the 2019 and 2024 modifications in criteria, as well as the historic drought of 2022, would strengthen our empirical analysis. Our study was unable to incorporate recognitions related to the 2022 drought due to an insufficient post-treatment period. This is particularly relevant given the centrality of the 2022 drought in the heightened awareness of the risks associated with the shrinkage and swelling of clays. A future study relying on property transaction data extending at least until 2025 would facilitate an analysis covering the entire period ruled by the Circular n°INTE1911312C 2019 criteria (2018-2023).

7 Concluding remarks

This study reveals that the impact of clay shrinkage and swelling on housing prices has only recently begun to manifest, though its influence remains modest. It offers critical insights into the operation and potential vulnerabilities of France's natural disaster compensation system. The findings suggest that while the Cat-Nat system has been effective in mitigating financial risks for homeowners, it may obscure market signals associated with these risks, leading to a possible underestimation of the true dangers posed by climate change. The quasi-experimental methodology employed in this research indicates that excluding shrink-swell risks from the insurance framework could significantly impact property markets, implying that the current system, though

effective, may be unsustainable in the long run without substantial reforms. Based on the damage projections generated by this study, the withdrawal of the Cat-Nat scheme could have catastrophic consequences, raising the question of whether the public-private system has become too big to fail.

Both academic (Bezy 2024) and institutional (Cour des comptes 2022, Sénat 2023) sources have highlighted that the Cat-Nat system, which spreads risk uniformly across the country, fails to provide disincentives through insurance premiums to deter building in high-risk areas. However, private insurance may create a “double-whammy” (Cannon et al. 2020): properties with insufficient or no insurance face both heightened risks and rising premiums in the context of climate change. A comprehensive evaluation of the sustainability of private insurance models — towards which an increasing number of governments are gravitating to address climate change — suggests that this approach may lead to a deadlock (Lucas and Booth 2020). The "privatization of climate adaptation," which shifts responsibility to individuals and aligns with state efforts to reduce the fiscal burden of escalating disasters, effectively frames climate change as a matter of personal responsibility. Our findings indicate that the risk of clay shrinkage and swelling is pervasive across income levels, while contributions to climate change increase with income (Chancel 2022). Consequently, Lucas and Booth 2020 argues that the response to natural disasters should be perceived as a collective responsibility.

The question of the optimal insurance regime to address climate change remains unresolved. Our study contributes to this literature by assessing the critical role of the Cat-Nat system’s palliative dimension, yet a portion of insurance surcharges funds an adaptation policy. In this context, the State’s policies for adapting to and preventing the risk of clay shrinkage and swelling, particularly through the Barnier Fund, merit further evaluation.

8 Bibliography

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9 Appendix

9.1 Shrink-swell: Predisposing factors and triggers

The increasing risk of clay shrink-swell due to the proliferation of droughts highlights a strong connection with climatic conditions. Nevertheless, the differential ground movement caused by drought and soil rehydration is a complex and multifactorial phenomenon, making it very difficult to predict its occurrence. Sophie Barthélémy's work (Barthelemy et al. 2024) stems from the observation that existing indicators poorly predict shrink-swell incidents and proposes a new drought index to improve the understanding of subsidence triggers.

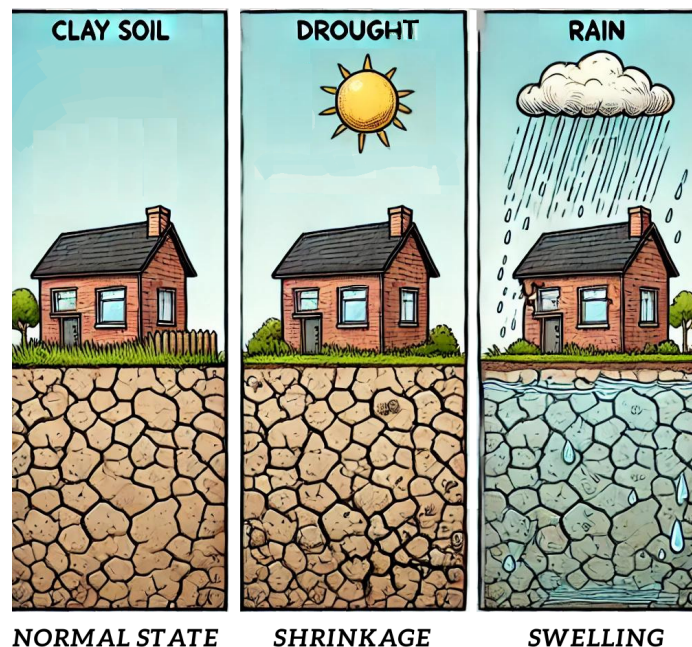
A set of **predisposing factors** help to inform the susceptibility of a territory to shrink-swell:

- Soil Nature: The proportion of clay material (in the granulometric sense, any fine element smaller than $2\ \mu m$), the mineralogical composition (some clay minerals are swelling, such as smectites or montmorillonites), and to a lesser extent, the geometry of the clay formation (a clay formation closer to the surface increases susceptibility).
- Hydrogeological Context: The presence of a water table partially protects against shrink-swell; its drying increases vulnerability.
- Vegetation: Trees near buildings cause water suction, which significantly increases susceptibility.
- Geomorphology: Topography and orientation influence the potential water reserve of soils. A sloping terrain can also increase the damage due to the asymmetry of the foundations it necessitates.

The presence of these predisposing factors makes the occurrence of incidents possible in the presence of **triggering factors**:

- Climatic Conditions: Seasonal variations normally change the water content at 1 or 2 meters, but an exceptional drought can affect depths of up to 3 to 5 meters. This explains why individual houses are threatened: their foundations rarely exceed 2m in depth, unlike taller buildings.
- Anthropic Factors: Development works such as drainage, pumping, waterproofing, as well as pipeline leaks and underground heat sources can also trigger the phenomenon.

Predisposing factors characterize the susceptibility of an environment, while triggering factors determine the occurrence of the phenomenon.



Source: Author's prompt, ChatGPT 4.0

Figure 7: Shrinkage and swelling of clay soils - a schematic overview

9.2 BRGM exposure map

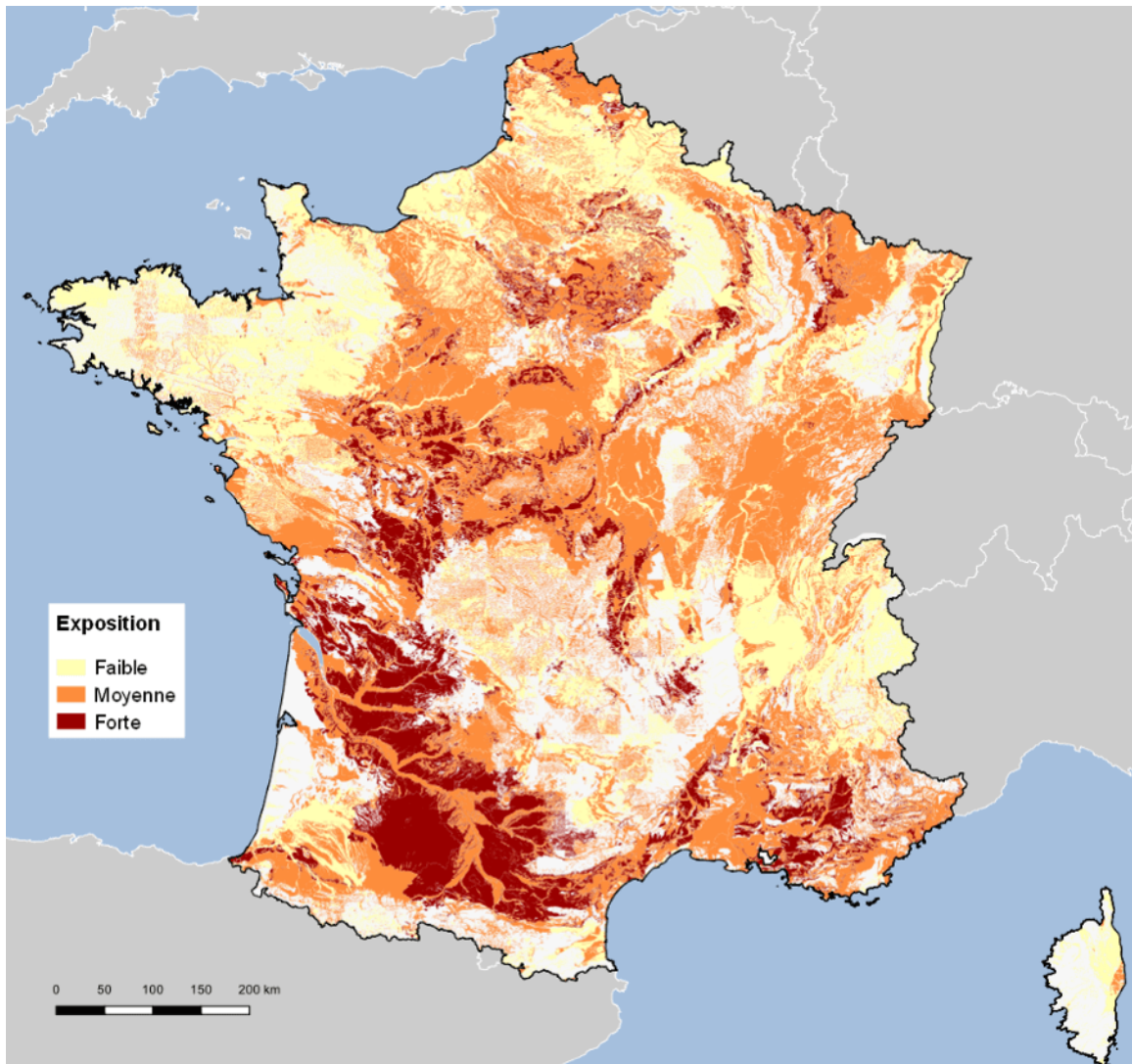


Figure 8: BRGM Exposure Map (ELAN law, 2020)

9.3 Hedonic model: list of controls

Spatial granularity	Covariates
House level	Built-up area, Plot area, Number of rooms, Age of the building, Number of levels, Number of bathrooms, Number of garages, Terrace, Pool, Other outbuildings (cellar, attic)
Iris level (around 2000 inhabitants)	Long-term unemployment rate, Share of population aged 30-39, Share of couples with children, Share of students, Interdecile ratio, Density of housing, Share of vacant dwellings, Share of secondary residence
Municipality level	Number of dwellings, Share of social housing
200m×200m grid	Distance to a train station, Distance to a metro or tram, Distance to a Seveso site, Distance to a priority urban area (QPV), Distance to the coastline, Noise pollution exposure plan, Distance to middle-school, Share of middle schoolers achieving honours

Table 6: Controls Used in the Analysis

Selected controls are listed in Table 6. As depicted in Figure 9, the chosen numerical covariates do not exhibit strong correlations. Categorical variables were excluded when their variance inflation factor (VIF) exceeded 10, as was the case for touristic infrastructures, which are highly correlated with the proportion of secondary residences in the Iris. The remaining covariates have VIF values below 3¹³, indicating that multicollinearity is not a concern.

Table 7 shows our baseline hedonic model, which explains 75% of the variance in house prices. Some economically relevant variables, such as energy efficiency or the presence of a lift, could not be added due to lack of access to data.

¹³Fixed effects have higher VIF values. Since the coefficients associated with our fixed effects are not the focus of this analysis, this does not pose a significant issue.

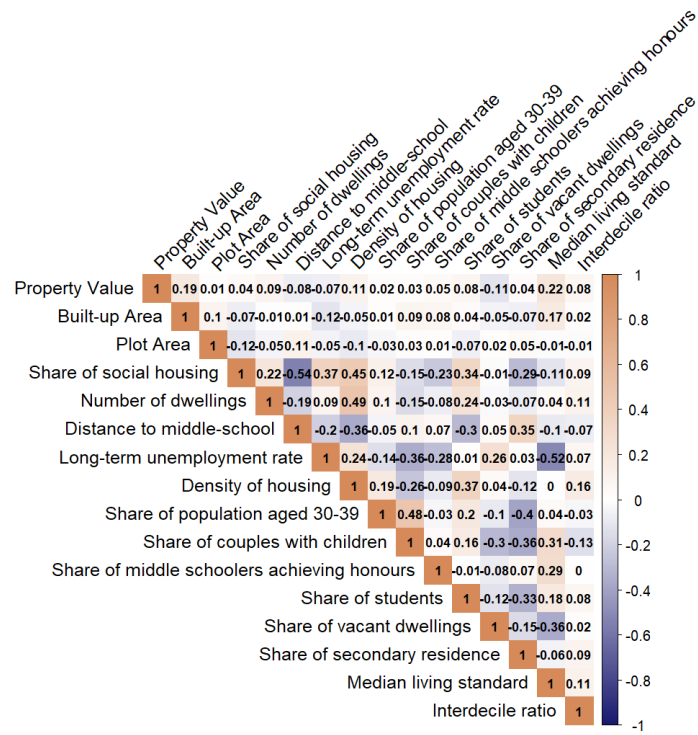


Figure 9: Multicollinearity check - correlation matrix of covariates (relevant house sales, 2010-2022)

Table 7: Hedonic model applied to French house prices (2010-2022)

Dependent Variable:	House prices (log)		
<i>Variables</i>		<i>Variables</i>	
Built-up area (log)	0.620*** (0.002)	Plot area (log)	0.049*** (0.001)
Three rooms or less	-0.039*** (0.001)	Five rooms	-0.009*** (0.001)
Six rooms	-0.016*** (0.001)	Seven rooms or more	-0.011*** (0.002)
Recent construction	0.042*** (0.001)	Built before 1914	-0.217*** (0.002)
Built 1914-1944	-0.168*** (0.003)	Built 1945-1960	-0.189*** (0.002)
Built 1961-1974	-0.147*** (0.002)	Built 1975-1989	-0.095*** (0.001)
Built 2013 and after	0.074*** (0.002)	Two levels or more	-0.048*** (0.001)
No bathroom	-0.180*** (0.002)	Two bathrooms or more	0.046*** (0.001)
No garage	-0.094*** (0.001)	Two garages or more	0.049*** (0.001)
Terrace	0.048*** (0.001)	Swimming pool	0.192*** (0.002)
Other dependency	0.053*** (0.001)	Long-term unemployment	-0.001 (0.001)
30-39 (share)	0.003*** (0.001)	With children (share)	-0.003*** (0.001)
Vacant housing (share)	-0.004*** (0.001)	Housing density	0.000*** (0.000)
Secondary residence (share)	0.002*** (0.000)	Decile ratio	0.078*** (0.008)
Median standard of living	0.000*** (0.000)	Proportion of students	0.001** (0.001)
Distance to middle-school	-0.004*** (0.001)	Middle-school honours (share)	0.001*** (0.000)
Station: <500m	-0.002 (0.002)	Station: <3km	0.023** (0.010)
Station: <5km	0.010 (0.009)	Station: <7km	0.005 (0.008)
Station: <10km	0.003 (0.006)	Metro/tram: <300m	0.043*** (0.008)
Metro/tram: <600m	0.034*** (0.006)	Metro/tram: <1km	0.023*** (0.005)
Seveso: <1km	0.002 (0.008)	Seveso: <10km	0.009* (0.005)
QPV	-0.064*** (0.004)	QPV: <500m	-0.014*** (0.003)
Noise exp. A	0.031 (0.022)	Noise exp. B	-0.042** (0.017)
Noise exp. C	-0.011 (0.007)	Noise exp. D	-0.024*** (0.008)
Coast: <300m	0.254*** (0.010)	Coast: <1km	0.119*** (0.007)
<i>Fixed-effects</i>		<i>Fit statistics</i>	
Year of transaction	Yes	Observations	5,232,345
Municipality	Yes	R ²	0.755
		Within R ²	0.495

Clustered (Municipality) standard-errors in parentheses

Significance Codes: ***: 0.01, **: 0.05, *: 0.1

Source: Metropolitan France house sales (2010-2022), DV3F; enriched by Insee PSAR-SL

9.4 Future exposure: list of figures

9.4.1 Geolocated projections at the house level

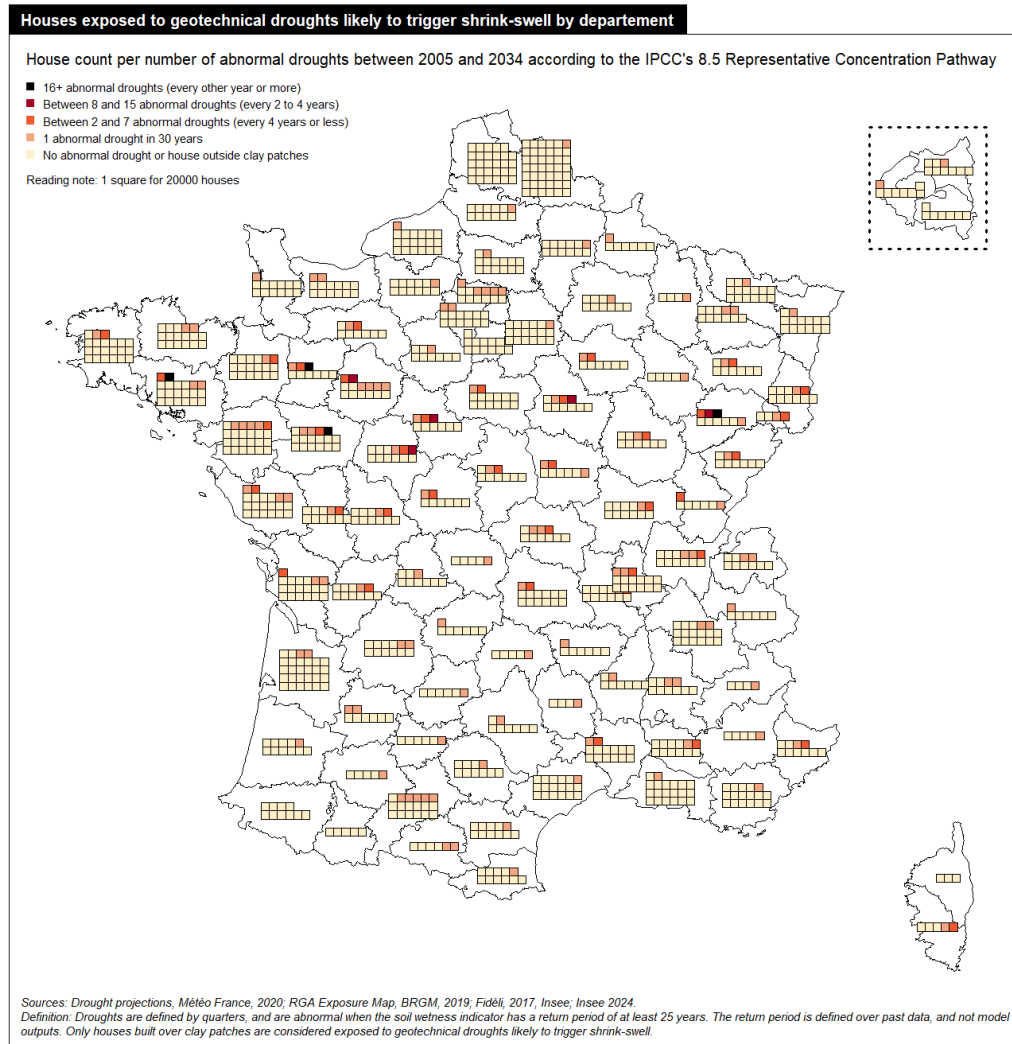


Figure 10: Shrink-swell damage projections per département - RCP 8.5 (2005-2034)

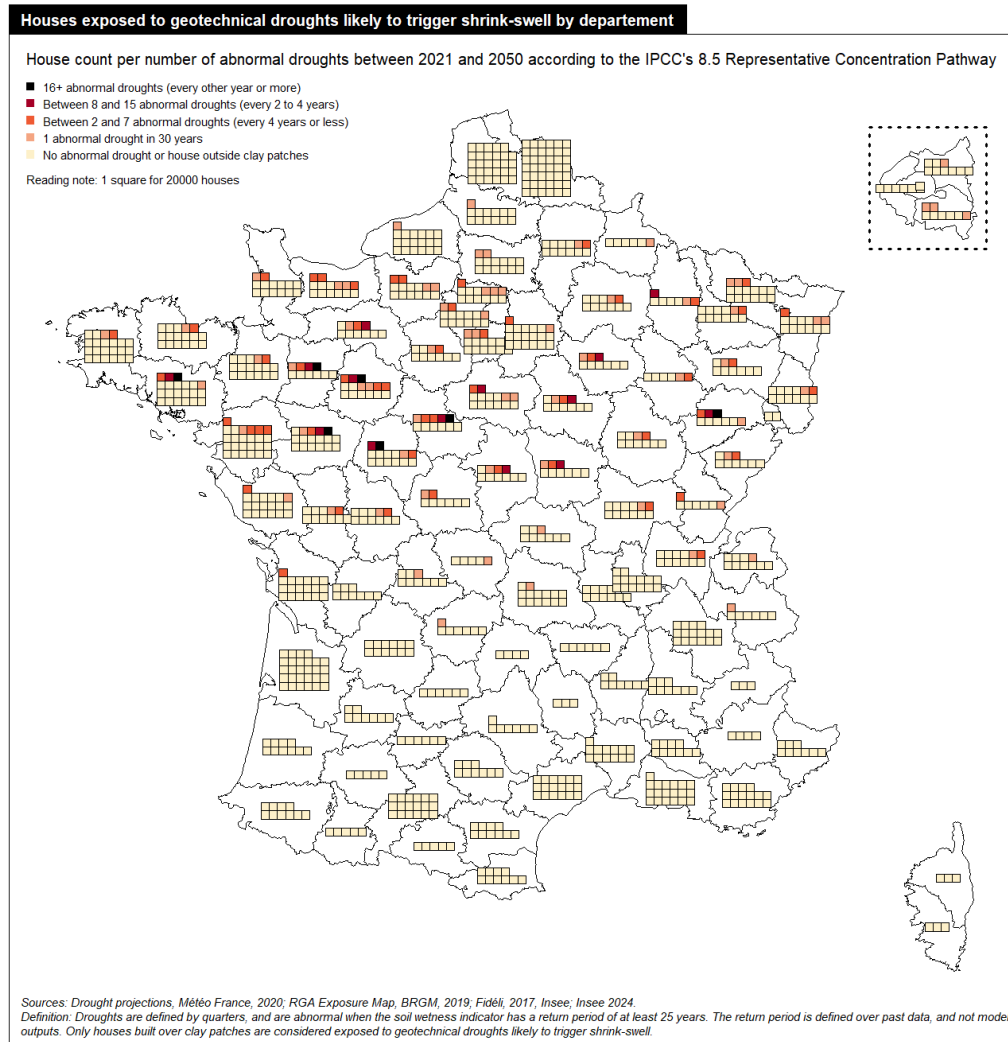


Figure 11: Shrink-swell damage projections per département - RCP 8.5 (2021-2050)

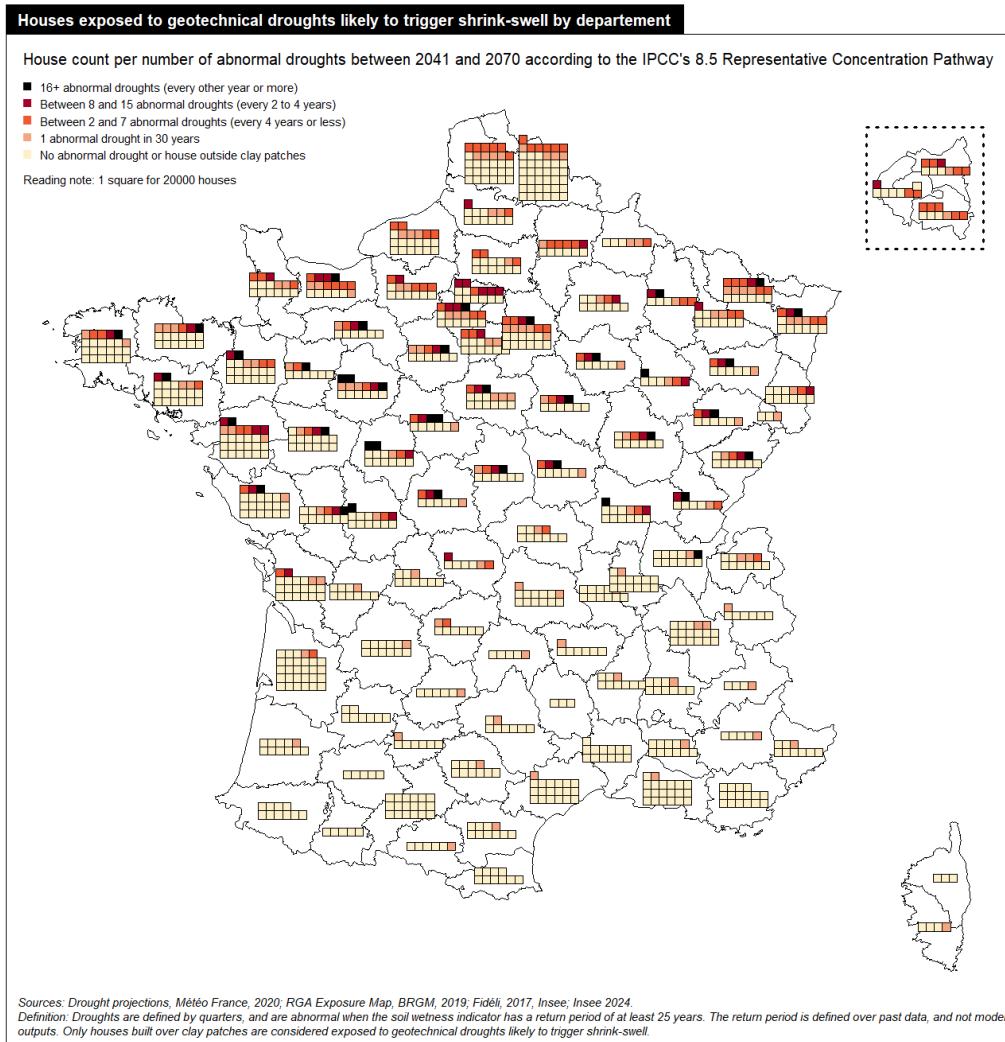


Figure 12: Shrink-swell damage projections per département - RCP 8.5 (2041-2070)

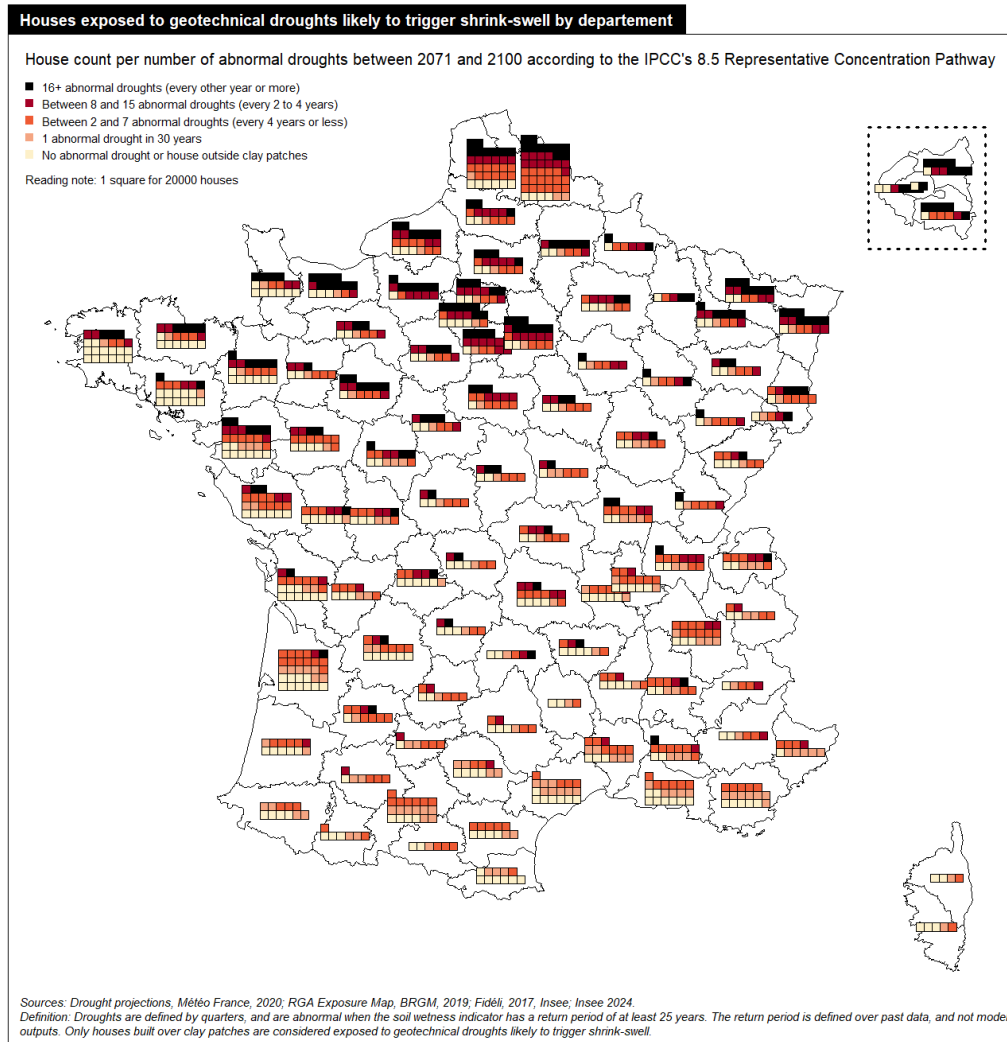


Figure 13: Shrink-swell damage projections per département - RCP 8.5 (2071-2100)

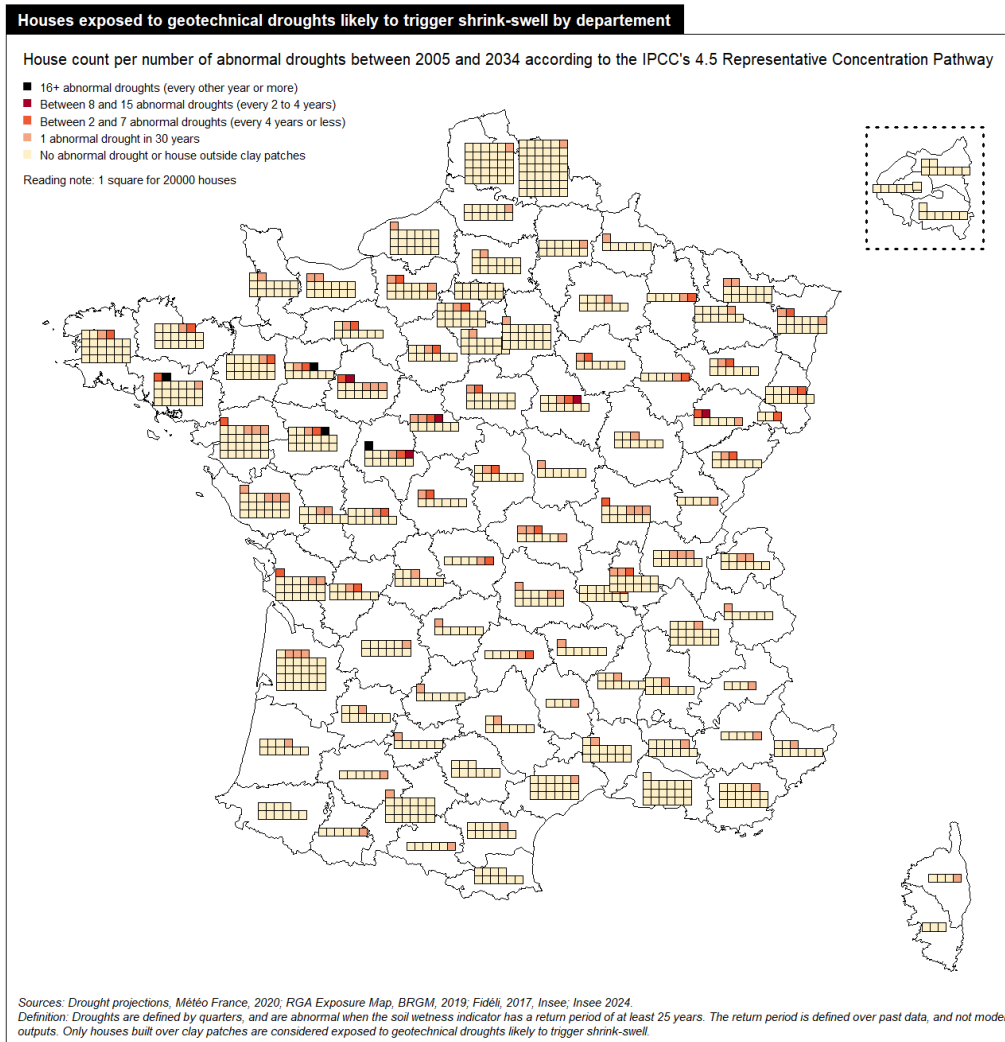


Figure 14: Shrink-swell damage projections per département - RCP 4.5 (2005-2034)

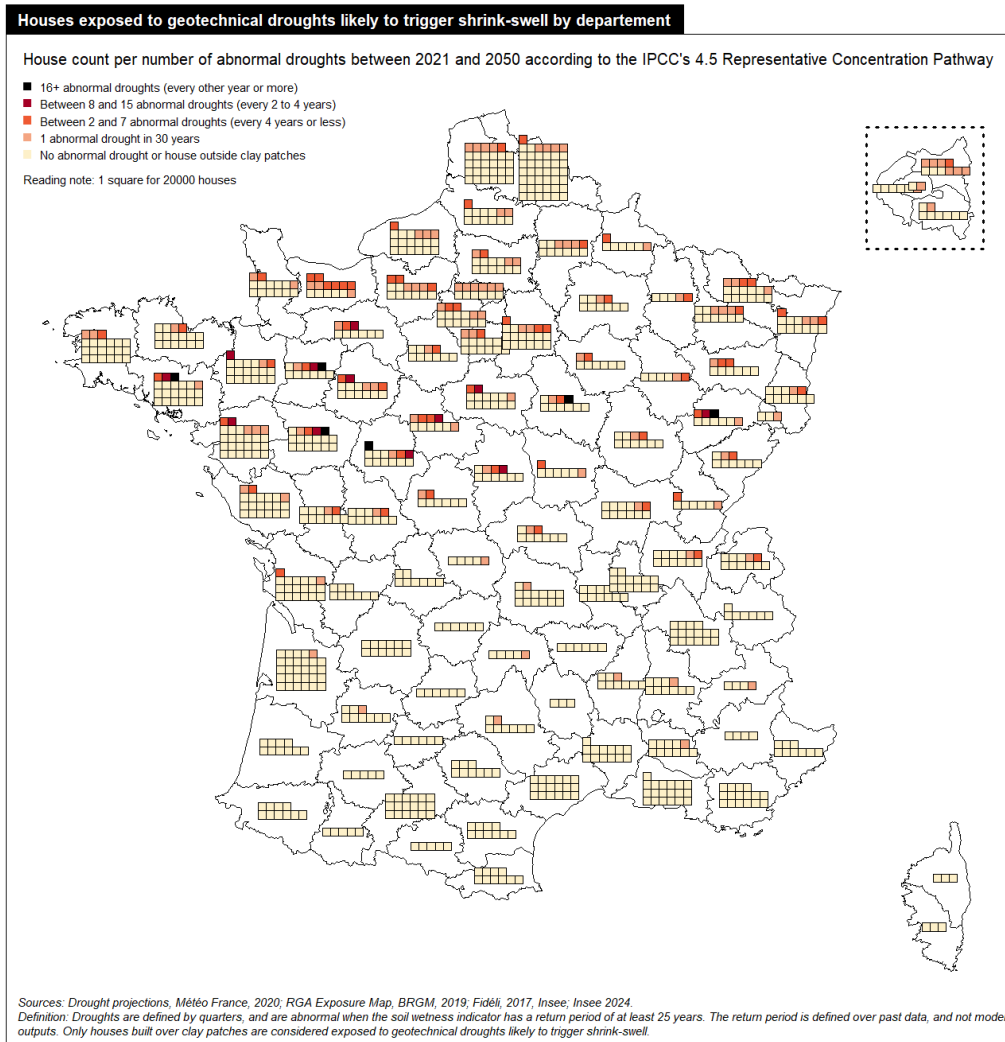


Figure 15: Shrink-swell damage projections per département - RCP 4.5 (2021-2050)

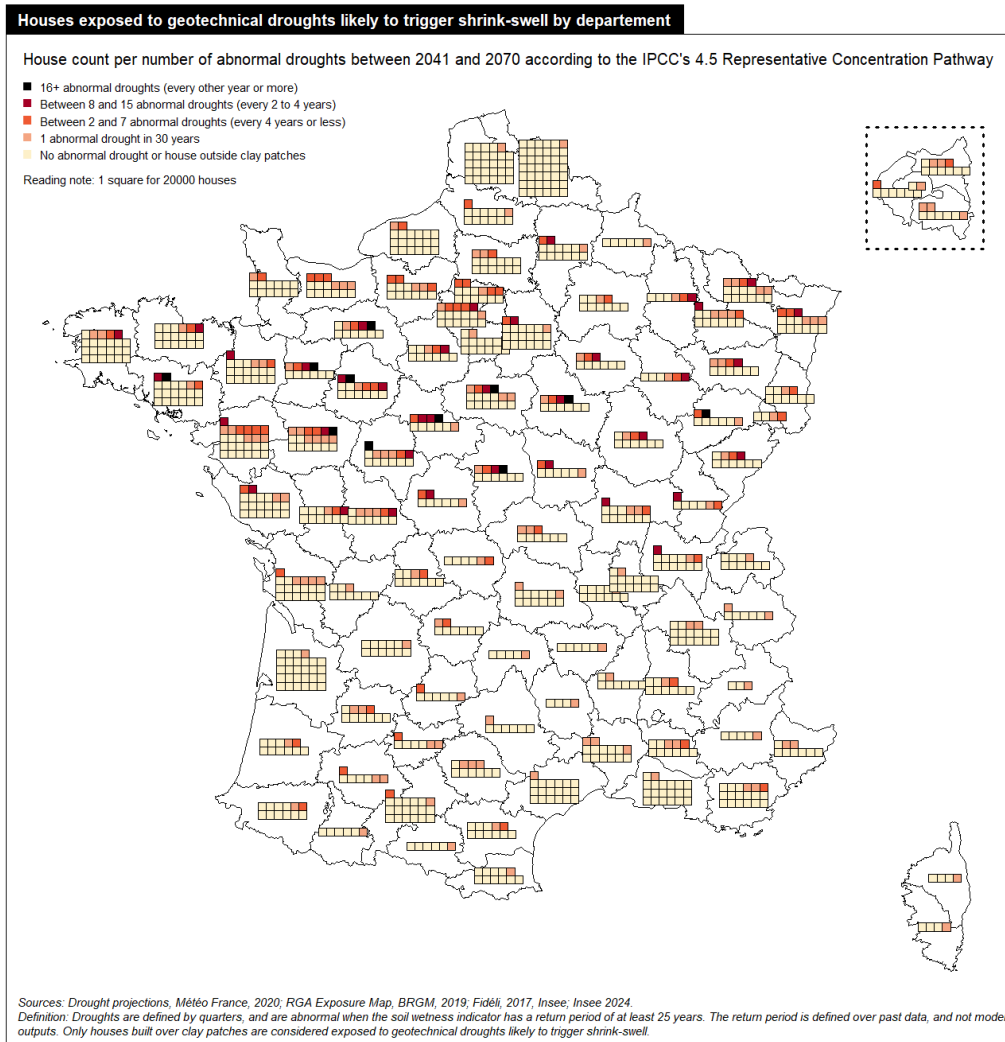


Figure 16: Shrink-swell damage projections per département - RCP 4.5 (2041-2070)

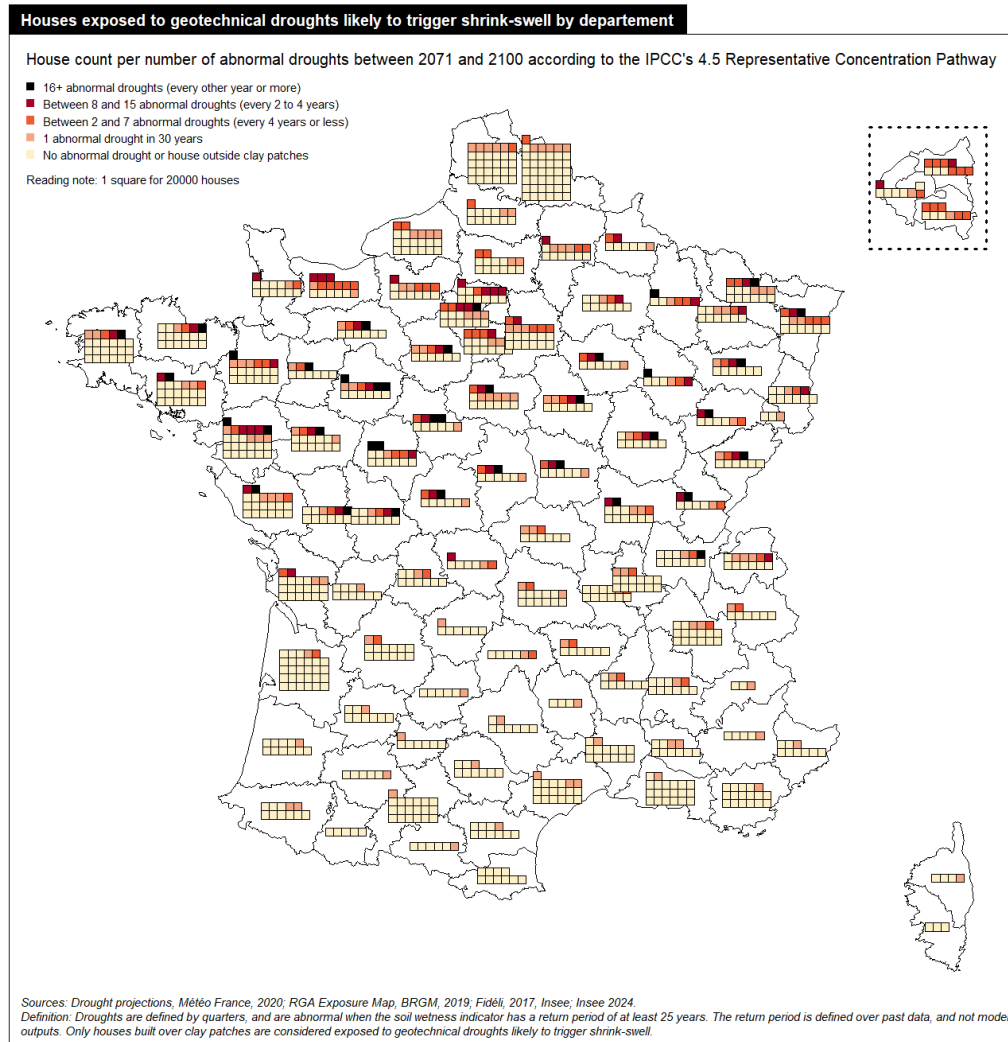
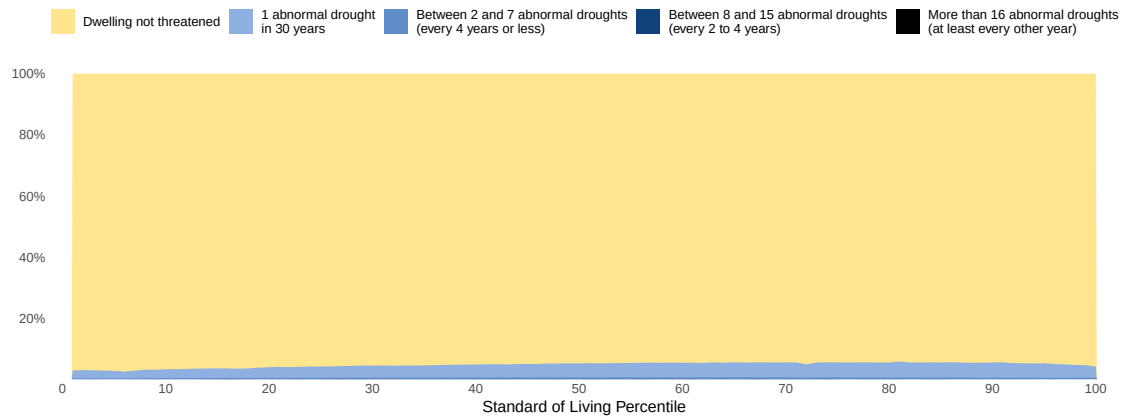


Figure 17: Shrink-swell damage projections per département - RCP 4.5 (2071-2100)

9.4.2 Living standard and shrink swell exposure

Households' Main Residence Threatened by Shrink-Swell in Metropolitan France between 2005 and 2034

Distribution of households by standard of living and frequency of geotechnical droughts likely to trigger shrink-swell and damage the main residence according to the IPCC's 8.5 Representative Concentration Pathway



Households' Main Residence Threatened by Shrink-Swell in Metropolitan France between 2021 and 2050

Distribution of households by standard of living and frequency of geotechnical droughts likely to trigger shrink-swell and damage the main residence according to the IPCC's 8.5 Representative Concentration Pathway

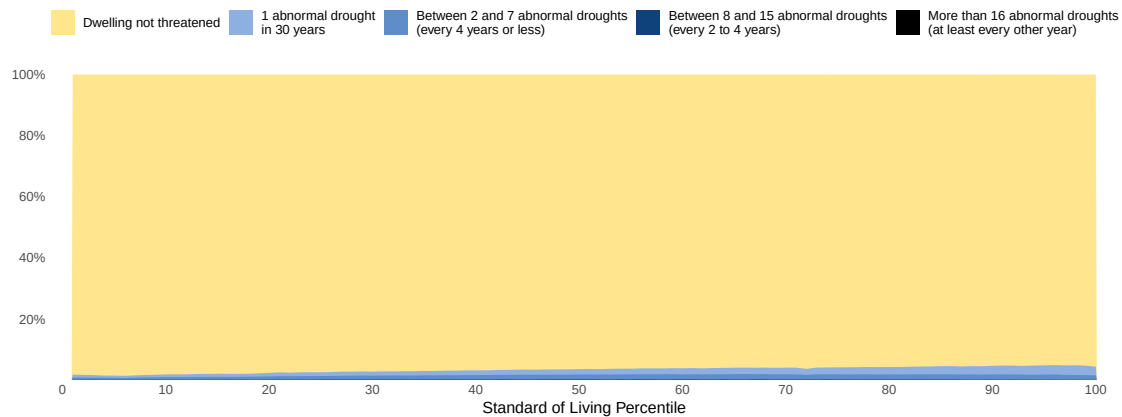
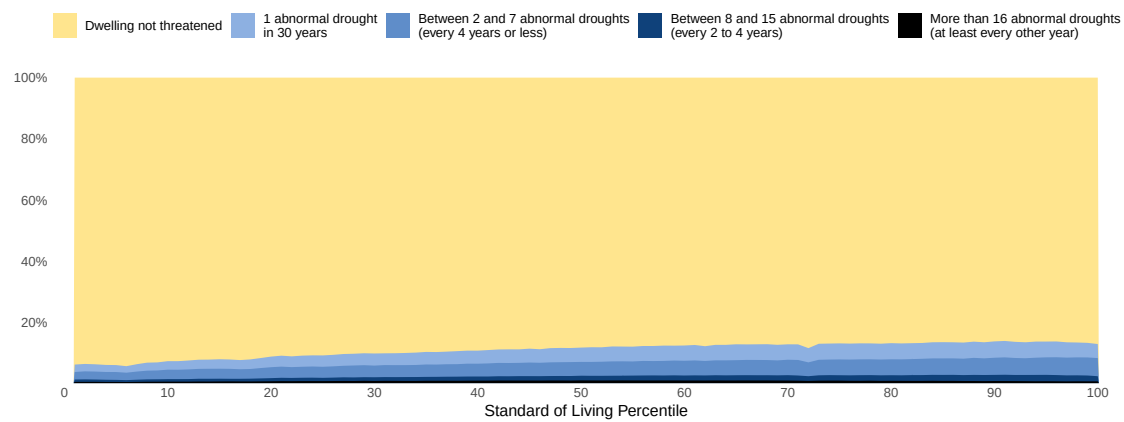


Figure 18: Climate justice - who'll be exposed to shrink swell ? (RCP 8.5, 2005-2050)

Households' Main Residence Threatened by Shrink-Swell in Metropolitan France between 2041 and 2070

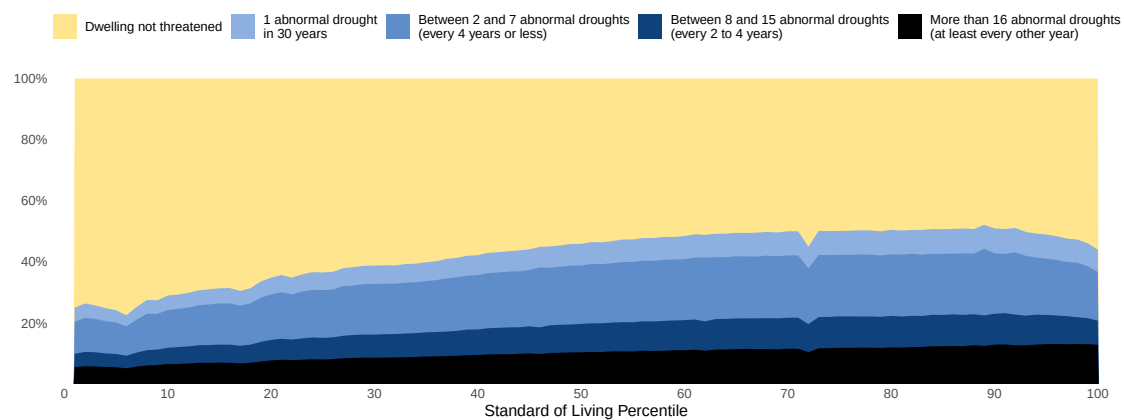
Distribution of households by standard of living and frequency of geotechnical droughts likely to trigger shrink-swell and damage the main residence according to the IPCC's 8.5 Representative Concentration Pathway



Dwellings not threatened by clay shrink-swell are houses located outside of clay areas, houses in low to high exposure zones without any geotechnical droughts with a return period of at least 25 years in 30 years, and all flats.
Sources: Clay Shrink-Swell Exposure Mapping, BRGM 2019; Soil Wetness Index projections, Météo France 2020; Fidéli 2017, Insee; Insee 2024

Households' Main Residence Threatened by Shrink-Swell in Metropolitan France between 2071 and 2100

Distribution of households by standard of living and frequency of geotechnical droughts likely to trigger shrink-swell and damage the main residence according to the IPCC's 8.5 Representative Concentration Pathway



Dwellings not threatened by clay shrink-swell are houses located outside of clay areas, houses in low to high exposure zones without any geotechnical droughts with a return period of at least 25 years in 30 years, and all flats.
Sources: Clay Shrink-Swell Exposure Mapping, BRGM 2019; Soil Wetness Index projections, Météo France 2020; Fidéli 2017, Insee; Insee 2024

Figure 19: Climate justice - who'll be exposed to shrink swell ? (RCP 8.5, 2041-2100)

9.5 Difference-in-difference results

9.5.1 2018-2019 Droughts

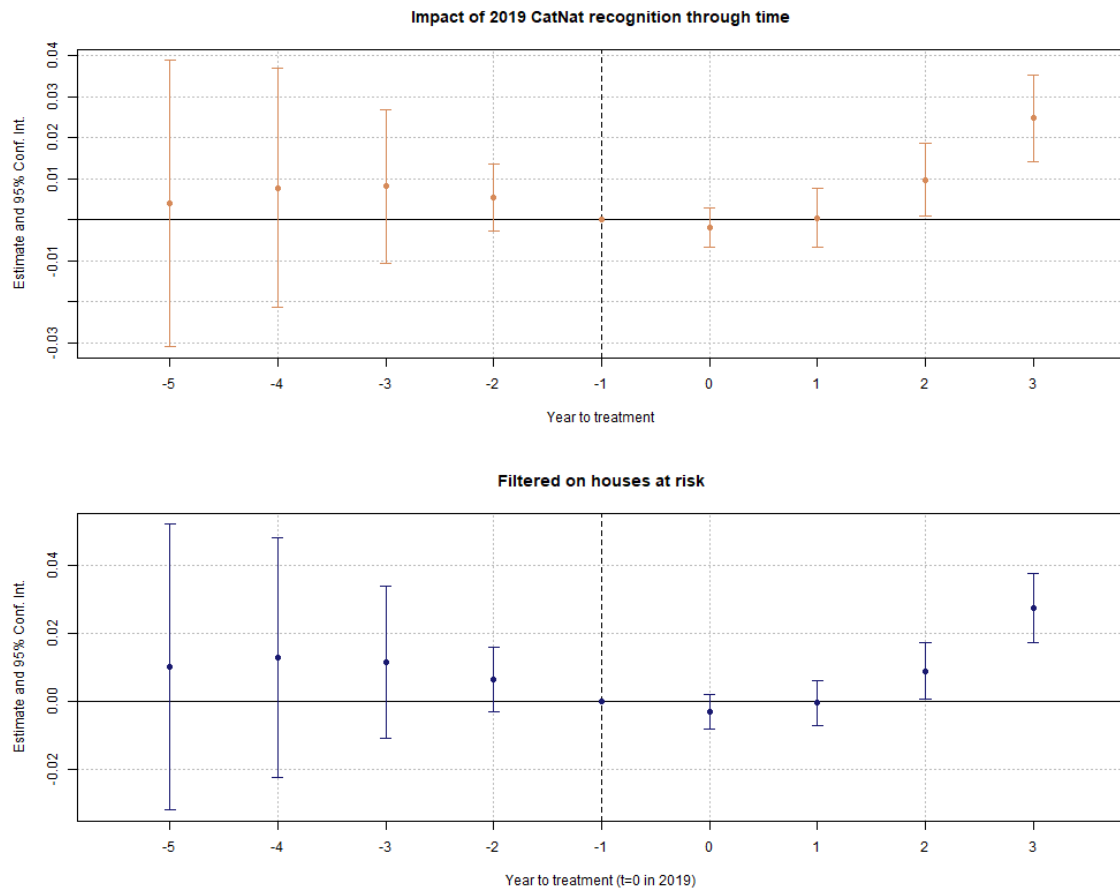


Figure 20: Assessing yearly treatment effects among **all municipalities** requesting Cat-Nat recognition in 2019 and over the **subset of houses exposed to shrink swell** - clustered standard errors

9.5.2 Full sample (stacked data)

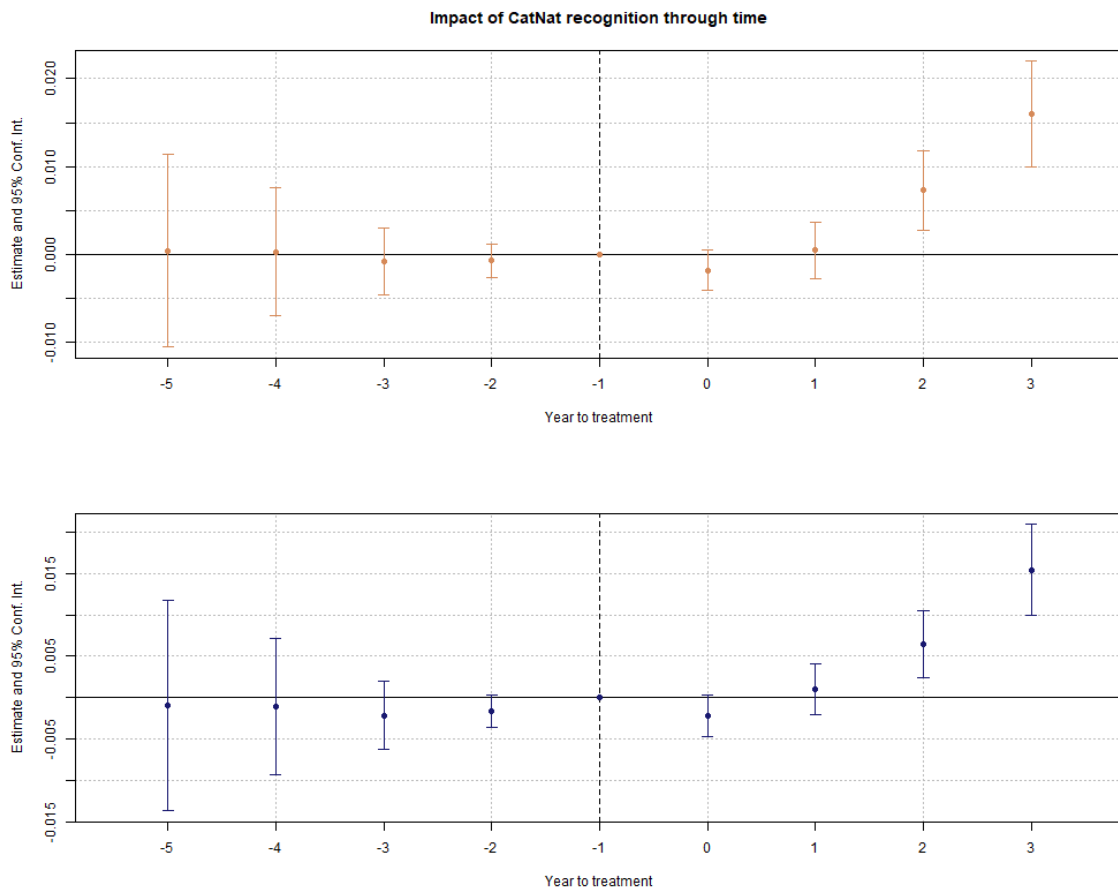


Figure 21: Assessing yearly treatment effects among **all municipalities** requesting Cat-Nat recognition in 2019 and over the **subset of houses exposed to shrink swell** - clustered standard errors

Table 8: Within-group pricing in French house prices (2010-2022) of shrink-swell risk
- all Cat-Nat assessments (2019-2022)

Dependent Variable:	House prices (log)	
Model:	(Among treated)	(Among untreated)
<i>Variables</i>		
Exposed $\times$ Years to treatment = -5	-0.018*** (0.005)	-0.009 (0.008)
Exposed $\times$ Years to treatment = -4	-0.014*** (0.004)	-0.014** (0.007)
Exposed $\times$ Years to treatment = -3	-0.009** (0.004)	-0.014** (0.005)
Exposed $\times$ Years to treatment = -2	-0.005 (0.004)	-0.010** (0.004)
Exposed $\times$ Years to treatment = -1	-0.002 (0.003)	-0.007 (0.005)
Exposed $\times$ Years to treatment = 0	-0.001 (0.003)	-0.005 (0.006)
Exposed $\times$ Years to treatment = 1	0.003 (0.004)	-0.002 (0.007)
Exposed $\times$ Years to treatment = 2	0.010*** (0.004)	0.004 (0.008)
Exposed $\times$ Years to treatment = 3	0.019*** (0.004)	0.020** (0.010)
<i>Controls</i>		
House characteristics	Yes	Yes
Neighbourhood characteristics	Yes	Yes
Local amenities	Yes	Yes
<i>Fixed-effects</i>		
Municipality	Yes	Yes
Year of transaction	Yes	Yes
<i>Fit statistics</i>		
Observations	1,676,105	691,271
R ²	0.79111	0.76341
Within R ²	0.50225	0.51102

Clustered (Municipality) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Source: Metropolitan France house sales (2010-2022), DV3F; enriched by Insee PSAR-SL; Shrink swell exposure maps (BRGM); Soil Wetness Index (Météo France)

9.6 Balance tests and robustness checks

9.6.1 2018-2019 Droughts

Table 9: Average Values per Treatment Status (Municipalities in the 2019 *Journal Officiel*)

Variable	Treated	Control
Property Value	222170	240830
Built-up Area	102.0	99.4
Plot Area	1150	976
Share of social housing	12.5	16.5
Number of dwellings in municipality	19300	13880
Distance to middle-school (km)	3.2	2.4
Long-term unemployment rate	4.3	4.5
Density of housing (/km2)	926	1010
Share of population aged 30-39	11.6	11.4
Share of couples with children	25.1	26.2
Share of middle schoolers achieving honours	66.2	64.8
Share of students	6.9	7.2
Share of vacant dwellings	7.9	7.9
Share of secondary residence	4.1	5.0
Median living standard (euros)	23030	23910
Interdecile ratio	2.9	3.0
Number of Municipalities	3086	475
Number of transactions	697760	119060
Number of unique transactions	695920	118840

Note: All differences are statistically significant (t-tests).

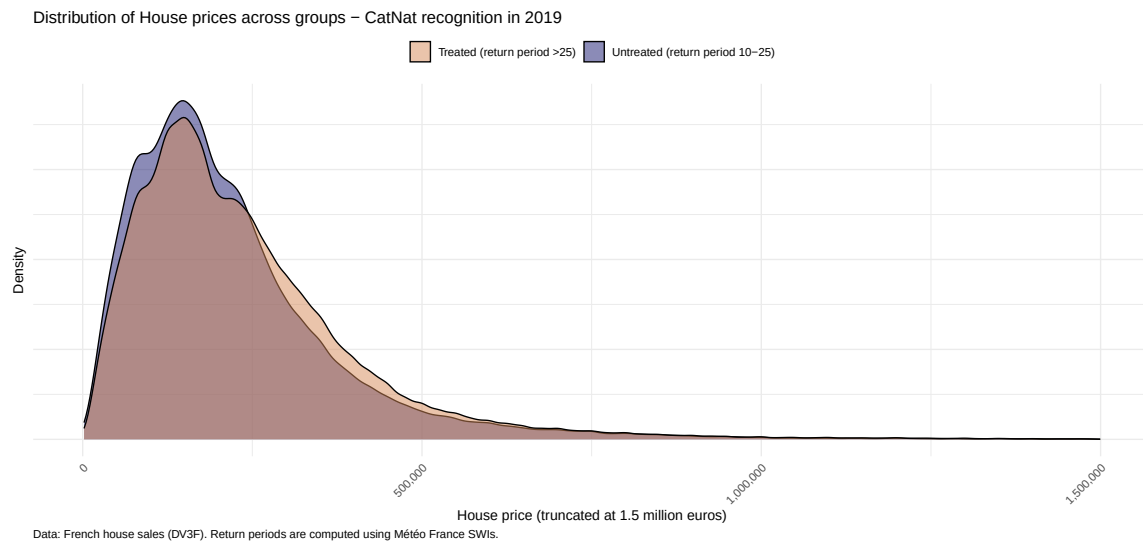


Figure 22: Distribution of prices across groups (Municipalities in the 2019 *Journal Officiel*)

9.6.2 Full sample (stacked data)

Table 10: Average Values per Treatment Status (Municipalities in any *Journal Officiel* 2019-2022)

Variable	Treated	Control
Property Value	243090	245820
Built-up Area	101.0	99.5
Plot Area	1010	979
Share of social housing	16.0	16.5
Number of dwellings in municipality	35260	20530
Distance to middle-school (km)	2.4	2.4
Long-term unemployment rate	4.5	4.7
Density of housing (/km2)	1110	1080
Share of population aged 30-39	11.7	11.4
Share of couples with children	25.7	25.8
Share of middle schoolers achieving honours	65.1	64.3
Share of students	7.5	7.2
Share of vacant dwellings	7.5	8.0
Share of secondary residence	3.6	5.0
Median living standard (euros)	23310	23500
Interdecile ratio	2.9	3.0
Number of Municipalities	5408	2330
Number of transactions	1680730	692910
Number of unique transactions	1133350	560700

Note: All differences are statistically significant (t-tests).

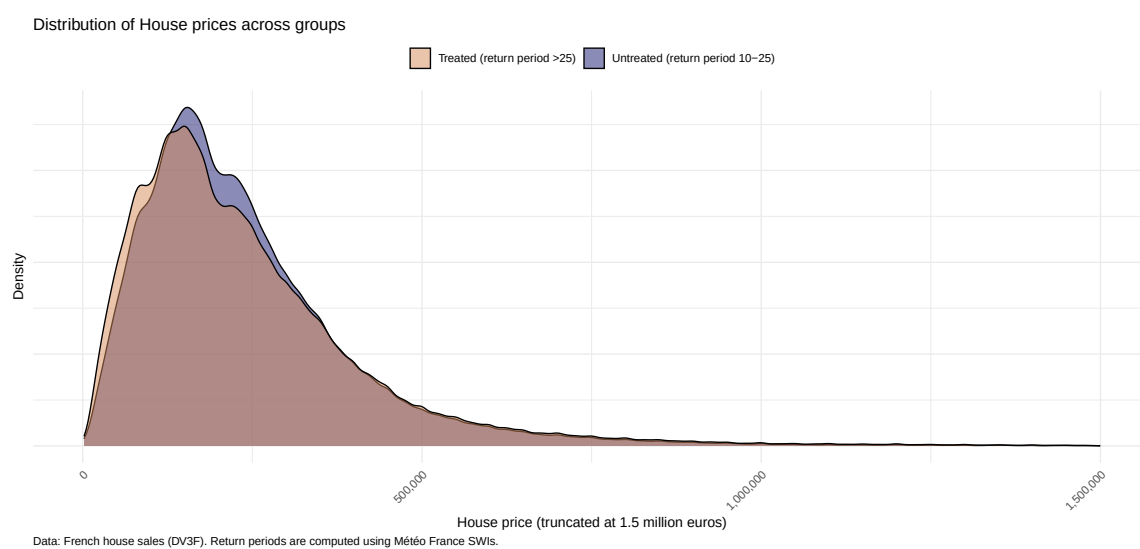


Figure 23: Distribution of prices across groups (Municipalities in any *Journal Officiel* 2019-2022)